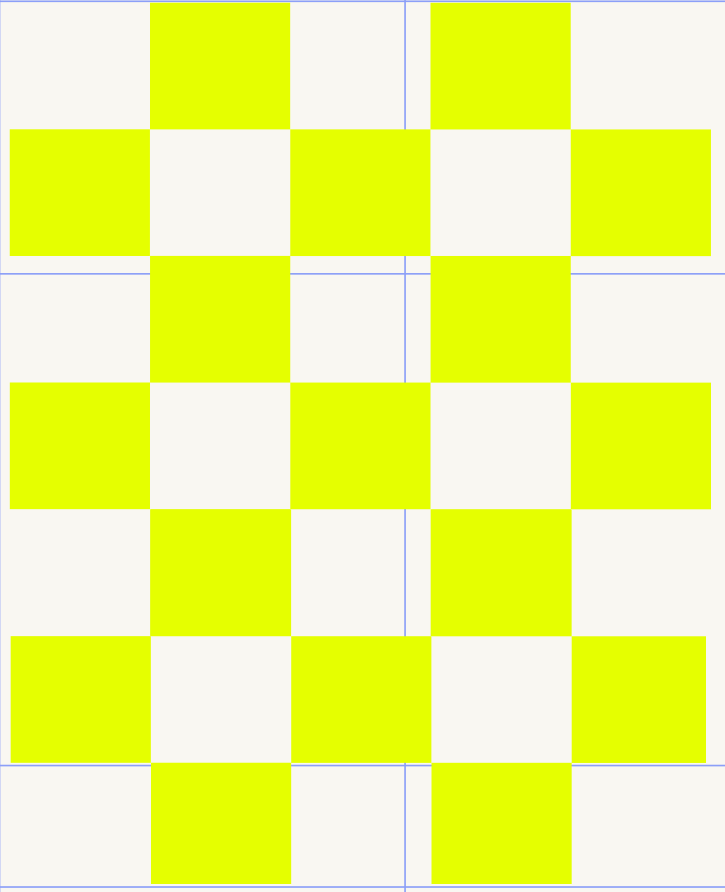
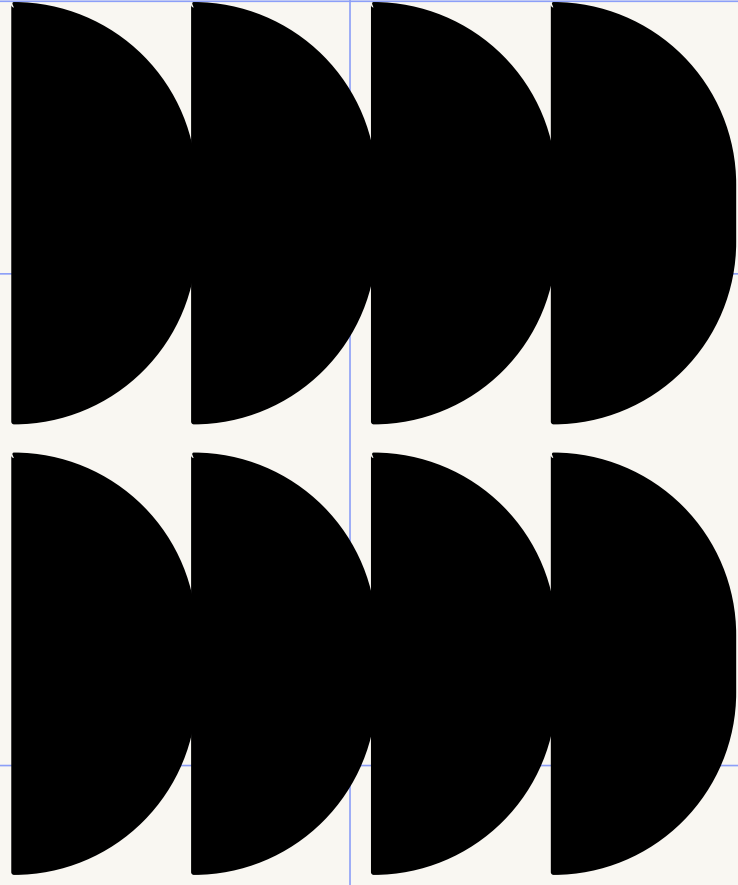




SHEHAN NIMSARA
S240253
S0SE

Design System Extended.

HCI PRACTIQ

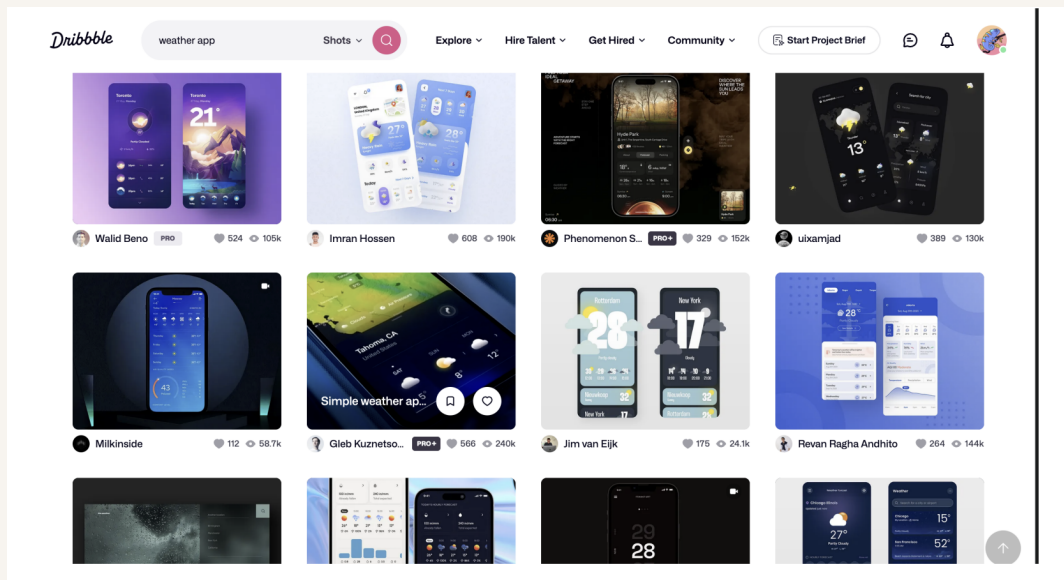
Table of Content

■	0.1	Immerse Yourself in HCI: Give and Take	3
■	0.2	Learn	5
■	1.0	Digital Prototype — Documentation 25%	10
	1.1	Design Process Reference	10
	1.2	User Research	10
	1.3	Design	13
■	2.0	Iterations 40%	19
	2.1	Laws of UX and Gestalt Principles	28
	2.2	UX Writing	29
	2.3	Accessibility Analysis (WCAG 2.2)	30
	2.4	User Testing	31
	2.5	Ad	32
■	3.0	The Physical Prototype 10%	33
■	4.0	Presentation 15%	36
■	5.0	Review 5% + 5%	38
■	6.0	Bonus Tasks & Appendix	42

0.1 Immerse Yourself in HCI: Give and Take

0.1 A) Consume

I spent about half an hour on Dribbble (dribbble.com) looking at weather app interfaces, interfaces that have to hold a lot of information in a small space and still be readable in about a second. That is the same problem any dashboard has, including a dashboard that matches people to jobs.



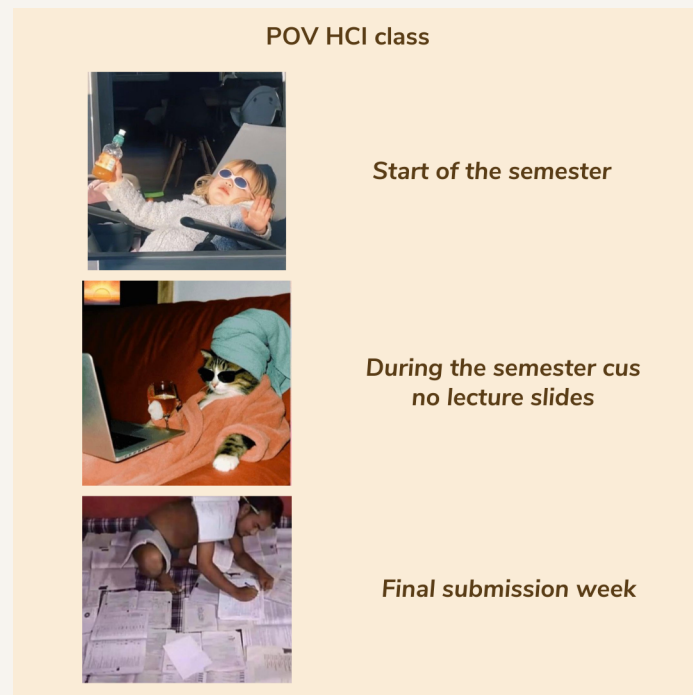
Dribbble search results for "weather app UI."

A few patterns kept coming up. The strongest designs are the ones that show almost nothing, only the current temperature, an icon for the condition and a short forecast. Colour is doing most of the work before you have read a single word, warm oranges for heat and blues or purples for rain and cold, so you have taken the reading before you decided to look. Temperature is nearly always the largest thing on the screen, often around 80–100 pt, which is what makes the hierarchy obvious without anybody explaining it. Almost all of them have a dark mode, which makes sense for an app that people open in bad light.

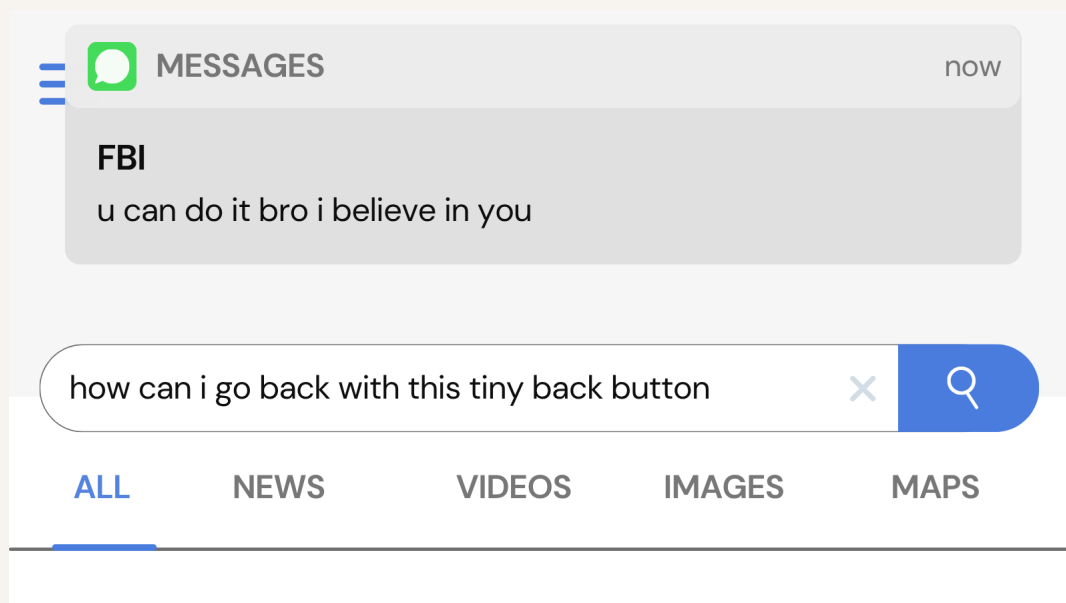
Before this I would have said that a weather app just shows you the weather. It does not. Type size, icon style, colour temperature and how much information sits on the screen are all decisions that somebody made on purpose. Not decoration that arrived by itself. I started seeing the same thing in my own screens about a day later.

0.1 B) Produce

For the produce half I made two HCI memes, memes that say the thing I would otherwise have written another paragraph about. It is closer to how I actually deal with a stressful module, and people pass a meme along in a way that they never pass a paragraph along.



“POV HCI class”: the semester arc from confident to underwater to buried in submissions.



A search-UI complaint disguised as a joke – the back button really is too small.

The first one is about workload across a semester, the gap between what somebody tells you in week 1 and what week 14 actually feels like. That is partly a mood and partly an information architecture problem, a problem about what the module tells you and when. The second one is a real complaint about interfaces wearing a joke on top of it. A back button with a tiny hit area fails exactly the target size rule that I later checked my own prototype against in §2.3. Getting an observation down into one frame turned out to be harder than writing a paragraph about it. I did not expect that.

0.1 C) Name and Analyse a Great Experience

Discord (discord.com) is the product I picked, and I use it daily because I am a gamer myself. It took a few minutes to get started. Join a server, find the channels, start talking. I have never once opened the help pages.

Dimension	Description
Name of the app	Discord
What does it do?	A real-time communication platform supporting text, voice and video channels. Users join “servers” organised into topic-based channels, used for gaming communities, study groups, open-source projects and general interest groups.
Target group	Primarily 16–30 year-olds with a strong overlap in gaming, tech and student communities; the UX is clearly optimised for younger, digitally native users comfortable with chat-centric interfaces.
Interaction modes	GUI (sidebars, channel lists, chat views on desktop and mobile); voice channels for hands-free real-time audio; slash commands (e.g. /giphy) bridging GUI and CLI interaction; notification-driven interaction via push and badge counts.

A few decisions explain most of why it works. The three column layout (server list → channel list → content) never moves, which means the mental model you build in the first ten minutes is still correct a year later. Everything that is interactive looks interactive, with clickable rows, hover states and tooltips, which is Norman’s affordance idea applied everywhere and not only in the places where somebody remembered. The advanced settings, the developer mode and the API keys and the audit logs, all sit a level or two further down, so a person opening the app for the first time never meets any of them. And there is feedback constantly, typing indicators and read receipts and voice status indicators, all of them telling you what is happening at this exact moment. That is what makes the place feel like other people are genuinely in the room.

What I took from it is that an app with a lot of features does not have to feel complicated. A consistent layout, clear affordances and progressive disclosure keep the cognitive load down no matter how much is underneath, and I used all three in the prototype work below and I also really like the color theme they used in it.

0.2 Learn

0.2 A) Learning Plan

Where I am starting from

I am in the 4th semester of a B.Sc. in Software Design at TH Aschaffenburg, specialising in data science, software engineering and systems design. The degree covers the technical side thoroughly. Everything I know about interface and visual design came from my own projects alongside it, and by now the interface work quietly falls to me in every group, even in the Requirements Engineering module and also in project management where we specifically had to design interfaces. I read visual work the way I read code.

What I could not do was work the way a designer works. Producing the artefact is one skill and owning the craft around it is another, meaning I could make the end product somehow but was not

an expert in designing using that too(Did not use the tool for its maximum) and I took this module partly to find out how far apart the two were for me.

The ambition

The ambition is to become an AI full stack engineer, an engineer who connects the far ends between the mathematics of AI and the User Interface that directly interacts with the users. A lot of AI work stops at the notebook, at a model that performs correctly and a person who never wants to touch it again. The best way I can describe what I want instead is as **a creative scientist**, a scientist who is also a designer. Not the person who only designs, or the person who only works with the logics. That is also why this module sits inside my degree and not beside it, because HCI owns the far end of that distance.

What I set out to learn

- **Front-end at studio level.** `lusion.co` is the benchmark — scroll-driven motion, WebGL, physics, interaction that is the idea itself. I planned the route bottom-up deliberately: React first (component model, props, state), then motion as a real material (Framer Motion, GSAP with Scroll-Trigger), then 3D through Three.js and React Three Fiber, and only then GLSL shaders. None of the upper layers are worth anything if the component model underneath is shaky, so I made myself not start at the shiny end.
- **Figma as a method, not a drawing surface.** Components and instances, variants, auto-layout, shared styles and tokens. Writing this down as a goal was uncomfortable, since it is precisely the thing people assume I already had.
- **A design position I can defend.** I knew what I liked and wanted to know why. Two traditions kept surfacing. Swiss / International Typographic Style gives me grid, hierarchy and type doing the work with nothing decorative. Neo-brutalism gives me raw edges, hard offset shadows and flat unapologetic hues with no gradient pretending to be depth. The working formulation I ended up with: Swiss gives the skeleton, brutalism gives the voice.
- **Typography with reasons behind it.** Getting from “this looks right” to being able to argue a three-voice hierarchy and defend it in a review.
- **The AI half, applied.** A model that determines how the screen is shaped and does real work, instead of sitting behind it unseen.

How I would know it had worked

I decided the evidence had to be artefacts, things that exist and can be opened, and not grades. A published portfolio built from scratch, a prototype with a design system that is actually systematic rather than drawn, and a deployed front-end that runs on state instead of being a click-through. All three exist now and all three are linked in this document.

0.2 B) Learning Journal

Framer, and shipping sheh4n.com. I learned Framer properly first and used it to build and publish sheh4n.com. The site taught me more than the tool did. It ended up with a CRT framed preloader counting up to 100 %, sections numbered 001–004 and metadata in bracketed monospace. That

is the same visual language that showed up in Practiq later, which I only noticed afterwards. Publishing makes you decide things a mock-up lets you put off forever. Breakpoints have to actually work, motion has to end somewhere, and every placeholder has to turn into a real sentence eventually.

Pinterest as a research habit. A lot of my visual input comes from saving layouts and palettes on Pinterest, and lime, purple and ink all started there. I think this only counts as research if the thing I take away is the rule and not the picture, and the rule here was “one saturated accent, one deep neutral, no third opinion,” a constraint that I could later defend on contrast grounds in 2.3. The board itself, together with the rule I took out of each reference, is in 2.0.2.

Learning Figma components halfway through. I started Practiq the way I had started every project before it, which is by drawing screens directly, one at a time, without ever thinking about how forty of them would be maintained later. Components and instances clicked somewhere in the middle of the project. A component is a single source and the instances inherit every change from it, which is completely obvious the moment somebody tells you and which reorganised how I saw the whole file. By then most of the variants already existed as hand-built copies, so the lesson arrived with almost nothing left to apply it to. The receipt for that is still sitting in the tablet build, where all six tablet match cards carry identical skill pills because they were duplicated instead of instanced, while the mobile cards, which were built later and built properly, carry the correct stack for each employer. Twenty minutes of documentation in week two would have saved every hour of it.

The design system arrived last. The v1.0 system page was pulled out of the finished screens rather than drawn before them. In 1.1 I argue that this order is defensible and I still think it is, but I should be honest that it began as not knowing the workflow. It only became a position afterwards.

Typography. I settled on three voices. Jomhuria for display, Metal as the warm italic accent, and JetBrains Mono for body copy, labels and data. I also spent time looking at German studio and agency work, including `pixeltypen.de`. That was not really about the typefaces. It was about how they handle a German and English voice at the same time without either one sounding translated, which is the same problem the DE build of Practiq had.

Dark mode and German as design work. I had both of these down as toggles in my head. Both turned out to be design work. For dark mode I had to rebuild the whole attention hierarchy around the lime accent, because once the shadows are gone there is nothing else directing the eye. There is no free inversion. German needed the same kind of attention. “Neues Match gefunden!” had to keep the warmth the English had, and every literal translation I tried flattened it.

The web build, and how it was actually made. I did not write the React from an empty file. I scaffolded the app with `Lovable`, pulled the repository down, and worked on it with `Claude Code` until it matched the design I had already made: the palette and the one-accent rule, the three type voices, the hard-shadow cards, EN/DE parity, the numbered overlay navigation, and a tracker where `ADVANCE` genuinely moves an application through its stages. What the generator produced was competent and completely generic, and nearly all the real work was in turning it into mine.

That is also where the bottleneck moved. When you are not the person typing the code, the scarce skill becomes specification, knowing exactly what you want in design terms and being able to tell when the thing that comes back is subtly wrong. Reading a diff and rejecting it turned out to be a real skill, a skill I did not know was a skill. The design system in 1.3.7 is what made rejecting anything possible at all, because it gave me something concrete to hold the output against. Motion

carries an accessibility cost as well, which is why prefers-reduced-motion is in the build.

The WCAG audit. Auditing the palette in 2.3 gave me the most useful rule I have now, which is that lime is never text. It is a surface or an accent shape with Ink on top of it. A rule I got to by measuring is worth more to me than one I got to by feel.

0.2 C) Learning Reflection

The short version is that I went from making pictures of systems to making systems, from drawing the thing to building the thing that generates it. Most of what is below is some version of that.

Knowing what a tool produces is not the same as knowing how to work inside it. I came into this able to make Figma files and I left able to build one, and nearly all of the learning sits in the gap between those two sentences. The cost is worth writing down rather than rounding off. I maintained 56 tablet frames by hand, and a set of match cards drifted out of sync without me noticing, because copies are copies and they inherit nothing. Next time the component set and the token sheet come before the second screen, not after the fortieth.

My design taste is a position now and not only a preference. Swiss structure with a brutalist surface used to be a thing that I liked on Pinterest without being able to say why. Now it is something I can point at inside the file itself, a spacing scale, four radii, one hard shadow at offset(-3,4), a strict one accent rule, and three named type voices with a defined job for each of them. Six months ago I could not have defended a single part of that in a review.

I am not yet where I said I wanted to be. Practiq's motion is CSS and JavaScript. No WebGL, no shader work, no physics. Measured against `lusion.co`, that is a lot of distance still to cover.

The React was AI assisted as well. I can read a component tree, direct changes into it and tell when the result that comes back is wrong. I have not yet built one on my own, and those are two different competences and not one. I would rather name both gaps than pretend that either of them is smaller than it is, especially since AI assisted building is the field I actually want to work in. Being vague about what I can do without it would be a strange place to start.

My next two steps are a small front-end from an empty folder with no generator involved, and then React Three Fiber with GSAP ScrollTrigger on something where the 3D is genuinely the point.

The gap that matters most is the AI one. Practiq's entire argument rests on the match score (96, 91, 88, 82, 79, 74), and those numbers are authored rather than computed. The interface is designed for a model that does not exist yet, which means **the score is currently a fiction**. That is uncomfortable to admit in a document that otherwise argues form follows function.

Making the number real is the clearest project in front of me now. Visa status as a hard constraint, CEFR level as a filter, and stack fit as a similarity score between a student's skills and what a posting asks for. Computed, ranked, and explainable enough that the interface can say why a listing got the score it got. It joins the two halves of what I want to be, and it would also show that the interface half was never the easy half.

How close Amir is to me, and why that is a risk. Amir is not a stranger to me. He is an international student in Germany with German sitting around A2-B1, a student who volunteers in the university's Buddy Programme helping other international students settle in. That closeness is the reason the FEELS quadrant in 1.2.3 came out as sharp as it did, and it is also the biggest methodological problem in the whole project. Designing for yourself feels exactly like research without being

research. Not a method, only a resemblance to one. That is why the user test in 2.4 matters as much as it does, because I need somebody who is not me to fail at these screens.

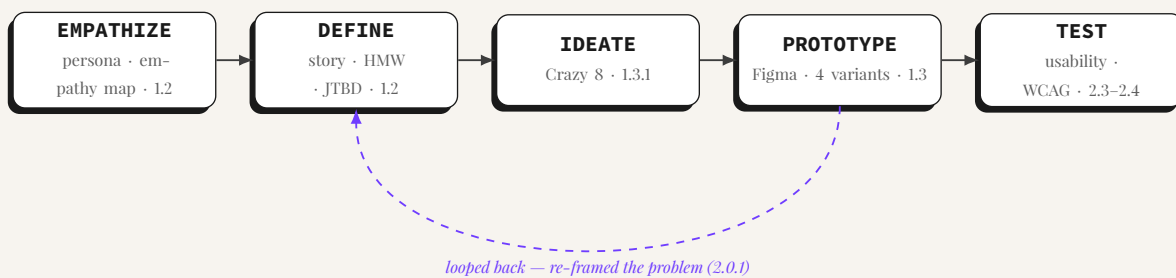
What I learned about how I learn. I learn by shipping things. The portfolio taught me more than the Framer tutorials, and the deployed web build taught me more than the Figma file. Every real lesson this semester happened when something had to survive an actual browser and an actual person. So I am going to keep picking the version of a task that ends in something published.

1.0 Digital Prototype — Documentation

1.1 Design Process Reference

The project ran on the Design Thinking model (Empathize, Define, Ideate, Prototype, Test), which continues the process I had already started in Portfolio Entry 1.1. I looked at Double Diamond and Service Design as the alternatives. Double Diamond is a model for a problem space that is still unclear, and after the empathy work my problem was already sharp, so what I needed was a process that stayed close to how the user feels while I was building. Service Design assumes value spread across many touchpoints and staff-facing back stages, and Practiq is not that kind of product. Design Thinking keeps a short line running from “what does Amir feel” to “what does this screen do about it,” and that line is exactly the thing I had to defend later.

In practice it was not a clean left to right pipeline, which is probably a good sign. I went through Empathize to Prototype once and then went back to Define. The first concept is in 2.0.1 and it was a plain ranked dashboard. The information on it was correct. But next to the empathy map it only answered half the problem, because it told Amir what existed and did nothing about how he felt while he was looking at it. So I went back and reframed the whole thing around emotion. The rest of the prototype comes out of that one decision.



You can see the process in the file and not just in this text. The Figma file still has the earlier generations of the six core screens sitting next to the current ones, so the iteration is there in the layers. The design system page is labelled v1.0 and I pulled it out of the finished screens. I did not draw a system first and then make the product obey it. That order is what **form follows function** actually meant here, in the sense the professor asked for. The feature and the flow came first, and the system is the pattern I noticed afterwards and then made consistent.

1.2 User Research

1.2.1 Target Group

Practiq is built for one group that nobody serves very well, international STEM students at German universities who have to complete a mandatory internship (the “Pflichtpraktikum”) inside a fixed semester window, and who are **non-EU nationals**. That last part is the whole point. Most German job boards are fine if you are an EU student. If you are not, there are two extra filters in front of every listing before anybody even talks about skills. Is this employer willing and legally able to take somebody on a student visa, and is the German level they want realistic? LinkedIn, Indeed and StepStone are built around how many listings they can show you, and none of them answer

those two questions. So this group ends up doing the filtering by hand, one tab at a time, while a graduation deadline gets closer. In Sinus-Milieu terms the group sits closest to the young, adaptive-pragmatic digital natives: mobile-first, efficiency-driven, and already fluent in the interaction patterns of consumer apps. The design leans on that last fact heavily.

1.2.2 Persona

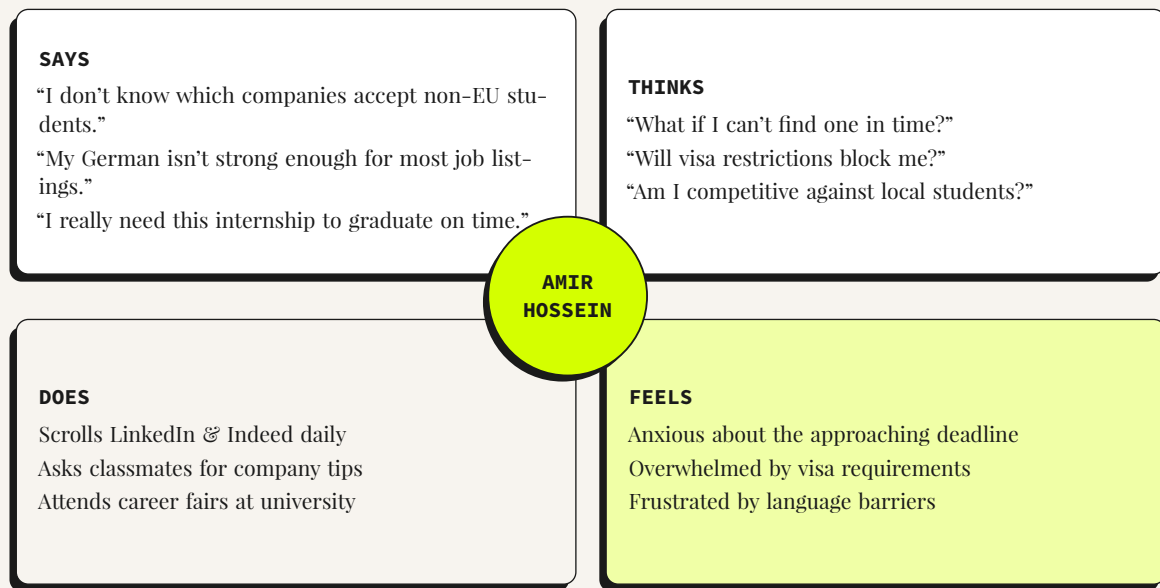
Amir Hossein, 23, Iranian national, in the 5th semester of a B.Sc. in Computer Science in Germany. His goal is simple and non-negotiable: finish the Pflichtpraktikum this semester so he stays on track to graduate. His problem is that he genuinely does not know which companies will take a non-EU applicant, his German is not strong enough for most postings, and he suspects he is being quietly out-competed by local students who do not have either barrier. Technically he is capable (Python, Java), but the friction is almost entirely outside the code.

Where Amir came from. Amir is not a composite assembled out of desk research and survey data. He is built directly out of what happened to me when I started looking for my own Pflichtpraktikum. Every friction in the paragraph above is one I hit myself: opening a listing that looked perfect and finding the German requirement three lines from the bottom, reading a posting twice before realising that “EU citizens or holders of a valid work permit” quietly excluded me, watching a deadline move closer while the shortlist did not grow, and losing whole evenings to a tab-by-tab sort that produced almost nothing worth applying to. The detail that Amir is technically capable and blocked almost entirely by things outside the code is not an insight I reached analytically. It is the part that annoyed me most.

That origin is a strength and a problem at the same time and I should say both. The strength is that it is precise. The FEELS quadrant in 1.2.3 is sharp because I was not guessing at any of it, and the eligibility-first ordering in 1.3.2 exists because I already knew which question I wanted answered before I would read a job description at all. You do not usually get a requirement that specific out of a survey.

The problem is that designing from your own experience feels like research even though it is not. It is a sample of one and it carries that one person’s assumptions without checking any of them, so the risk is that I solved my own problem and then decided it was everybody’s. I widened it where I could, against the Sinus-Milieu profile in 1.2.1 and against other international students I work with in the Buddy Programme, who told me about frictions I had not hit myself. I come back to this in 0.2 C. It is also why the user test in 2.4 matters more here than it would on a project I had no personal stake in.

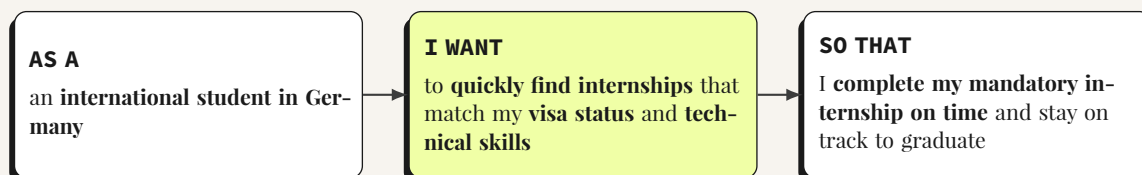
1.2.3 Empathy Map



The FEELS quadrant (highlighted) is the one I treated as a requirement, not a by-product — it is the brief the rest of the design answers.

I used this map in a specific way and it is why the app looks the way it does. Most job apps would treat SAYS and DOES as the requirements and stop there. I treated FEELS as a requirement too. "Anxious about the approaching deadline" is not something I can decorate away. It is a state that the interface either makes worse or makes better every time Amir opens it. Every gamified thing in 1.3 and 2.1 goes back to a specific line in this quadrant. If it did not go back to one, it did not go in the app.

1.2.4 User Story



1.2.5 Jobs To Be Done

- **Filter out the impossible before spending effort.** Amir needs to know a listing is compatible with his visa and language level "before" he invests an hour writing an application, not after a rejection tells him so.
- **See that he is actually getting somewhere.** A flat list of listings does not answer his real underlying question, which is "am I making progress against the deadline?" He needs visible, accumulating proof of movement.
- **Get back a sense of control.** Visa acceptance and graduation timing feel externally decided, which is where most of the anxiety comes from. The product should hand him back the parts he "can" influence and make them feel like progress.

1.3 Design

1.3.1 Ideation — ideas and why

I framed the design space with a How-Might-We question, trying to keep it neither too narrow, which would pre-decide the solution, nor too broad, which would be unactionable:

HOW MIGHT WE

help international students in Germany **efficiently discover and apply** for internships **compatible with their visa status, language level and technical background** — without manually filtering hundreds of irrelevant listings?

visa status

language level

technical background

A Crazy 8 round gave me eight directions: (1) Filtered Internship Portal, (2) AI Chatbot Assistant, (3) Smart Matching Algorithm, (4) Alumni Mentor Network, (5) Application Deadline Tracker, (6) CV Localisation Tool, (7) Company Review Board, (8) Multi-Platform Aggregator. The time pressure helped because it pushed me past the obvious first answer, which was to build another job board, and into things like the alumni network and the review board that I would not have got to otherwise. I carried forward 1, 3 and 8. Between them they go after the one thing underneath all three FEELS lines, which is discovery friction. If the right listings that he can actually apply for come up on their own, then the anxiety and the overwhelm and the feeling of being out-competed all drop at once.

Form follows function. I designed a feature before I designed any screen. The feature is the eligibility match, which is visa status $\times$ language level $\times$ tech stack collapsed into a single number. Everything you can see in the final UI comes after that number and is there to serve it. The ranked match cards present it (96, 91, 88, 82, 79, 74), the filter tabs let Amir narrow it (All · Visa ok · B1 Friendly), the home notification celebrates a new one (“New Match Found! 92% skill match”), and the XP system rewards him for acting on it. The visual layout was the last thing I decided.

1.3.2 Design Patterns & Information Architecture

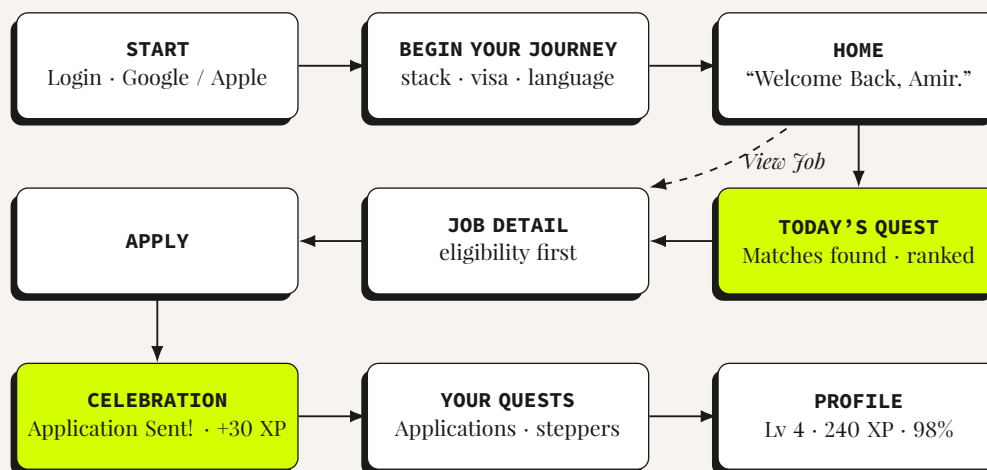
The information architecture is shallow on purpose and it loops back on itself, because Amir is a user who comes back to check status far more often than he starts a new search. The two screens he actually lives in are the match list and his applications, and those two are always one tap apart.

- **Dashboard home** (“Welcome Back, Amir.”) leads with a single New-Match notification card and then four colour-coded menu cards: Jobs (purple), Applications (lime), Profile (white) and Stack (black). The colour coding means each destination is recognisable before the label is even read, which quietly helps a user whose German reading speed is a stated barrier.
- **A quest-framed, ranked match list** (“Today’s quest — Matches found”) is the discovery surface. Matches are cards ordered by score (96 · 91 · 88 · 82 · 79 · 74), tagged with “Rare” where relevant, and narrowed by three filter tabs (All · Visa ok · B1 Friendly). I chose ranking over an open, unsorted board deliberately: the ranking answers “which one first?” before Amir has to spend energy asking it, and the filters answer “can I even apply?” in one tap, so he reads from the top instead of scanning.
- **A four-slot bottom navigator** (Home · Matches · Applications · Profile) carries the three screens Amir browses in — Matches, Applications and Profile. Home is the hub it returns to. The two

screens built around a single decision, Job Detail and the Celebration, drop the bar deliberately so nothing competes with the one action on them.

- **Sidebar navigation on tablet**, carrying the same four destinations plus the theme and language toggles and Amir’s avatar. I did not simply stretch the phone layout, because on a landscape screen a side rail is the more ergonomic reach. The tablet Home goes further and drops the rail altogether for a two-by-two grid of the menu cards with a Recent Activity feed underneath, content the phone has no room for at all. That is the clearest evidence that the tablet build is a genuine re-layout.
- **Progressive disclosure on Job Detail**. The screen answers Amir’s fears in the order he feels them. The purple header names the employer, the role and where it sits (Mandatory Internship · Berlin : Remote), the skill pills (Python · AWS · Kotlin · CSS) sit directly beneath it, and only then come About the Role, Requirements and What You’ll Get. He gets the “can I?” answer before the “should I?” details ask for his attention.
- **A gamification layer as the emotional spine**. This is the layer that separates Practiq from a normal job board and it is the direct answer to the FEELS quadrant. Applying earns +30 XP, the Profile shows Level 4, a 7-day streak and 98% Profile Strength, strong matches wear a “Rare” badge, and the whole journey is framed as quests: matches are “Today’s quest,” running applications are “Your quests” with a four-step progress bar (Applied → In review → Interview → Decision) and countdown badges (“4D left”). The detail I am most proud of is that a rejected application is never marked as failed. It reads **Quest complete**, which is a deliberate move against the empathy map’s “Am I competitive?” line and something I return to in 2.1.

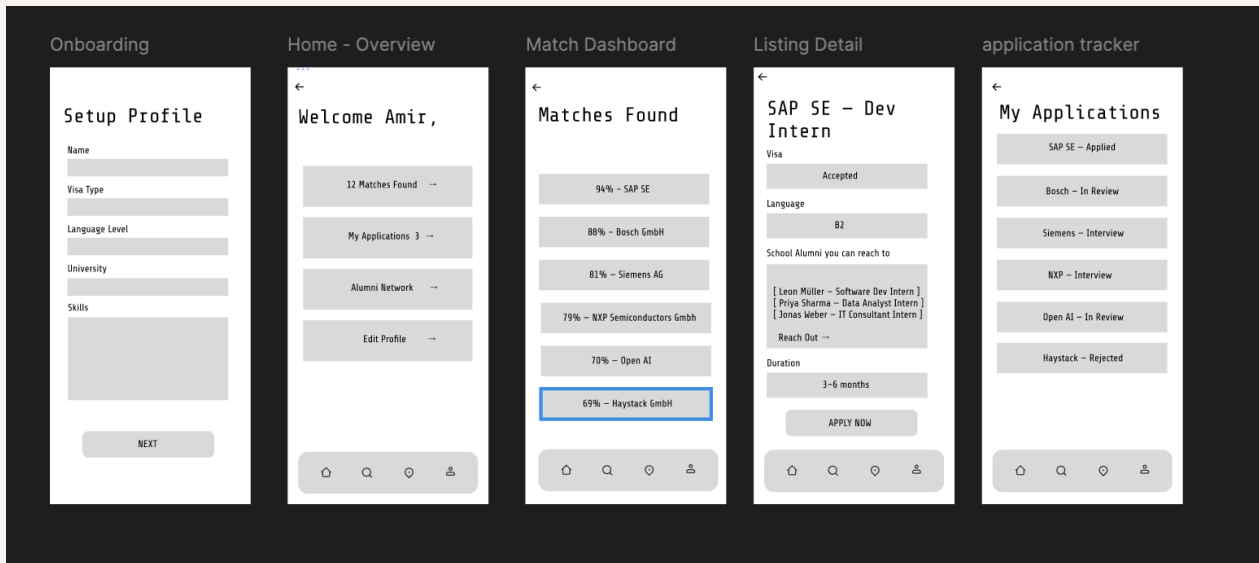
1.3.3 Flowchart — the possible journeys



Secondary journeys: Profile → edit Skills, the Home notification card jumps straight to Job Detail (“View Job”, dashed), and Logout from Profile.

1.3.4 Wireframe

The low-fidelity wireframes come from Portfolio Entry 1.1: Onboarding → Home Overview → Match Dashboard → Listing Detail → Tracker. No colour, no styling. Keeping them grey forced me to argue about structure and order before I could hide behind visuals, and that is where the information architecture got decided.

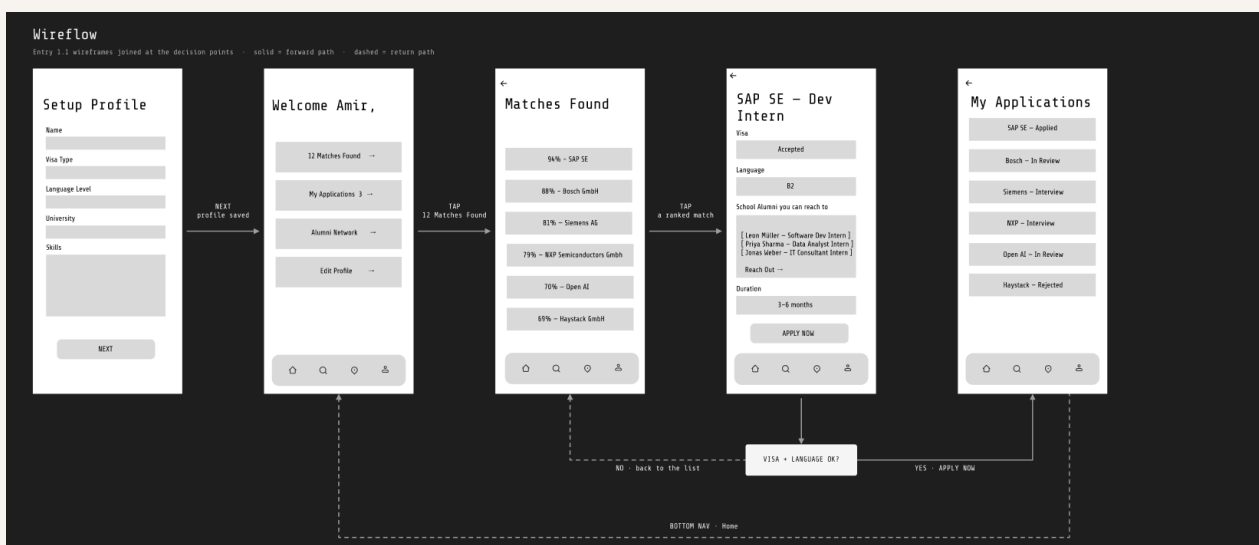


The five low-fidelity wireframes from Entry 1.1 — grey, unstyled, and deliberately ugly, so that the only thing left to argue about was order.

Two things in this sheet did not survive, and I would rather name both than quietly drop them. The employers and scores were placeholders (94 SAP, 88 Bosch, 81 Siemens) later replaced with the seven real companies. And Listing Detail carried an **Alumni Network** block offering named alumni to reach out to. I cut the alumni feature because it was a second product wearing the first one's clothes: it needed its own data, its own trust model and its own moderation, and none of that was answering the discovery friction the rest of the app exists to remove.

1.3.5 Wireflow

The Entry 1.1 wireflow joined those five wireframes at the points where the user has to decide something, and it is the skeleton of the finished product. You can map it one to one. Onboarding became Start plus “Begin your journey,” the Dashboard split into Home and the Matches list, the one Listing Detail turned into the per-employer Job Detail screens, and the Tracker became “Your quests.”



The wireflow — the same five wireframes joined at the points where Amir actually has to decide something. Solid arrows carry the forward path, dashed arrows carry the returns, and the one box that is not a screen is the decision the whole product was later built to remove.

That one decision node is the whole argument in small. In Entry 1.1 Amir had to open a listing before he could find out whether his visa and his German were good enough, so the flow forks under Listing Detail. Yes sends him to APPLY NOW. No sends him back to the list with the effort wasted. The eligibility score, the Visa ok filter and collecting the context once at onboarding are all there to delete that fork before he ever gets to it.

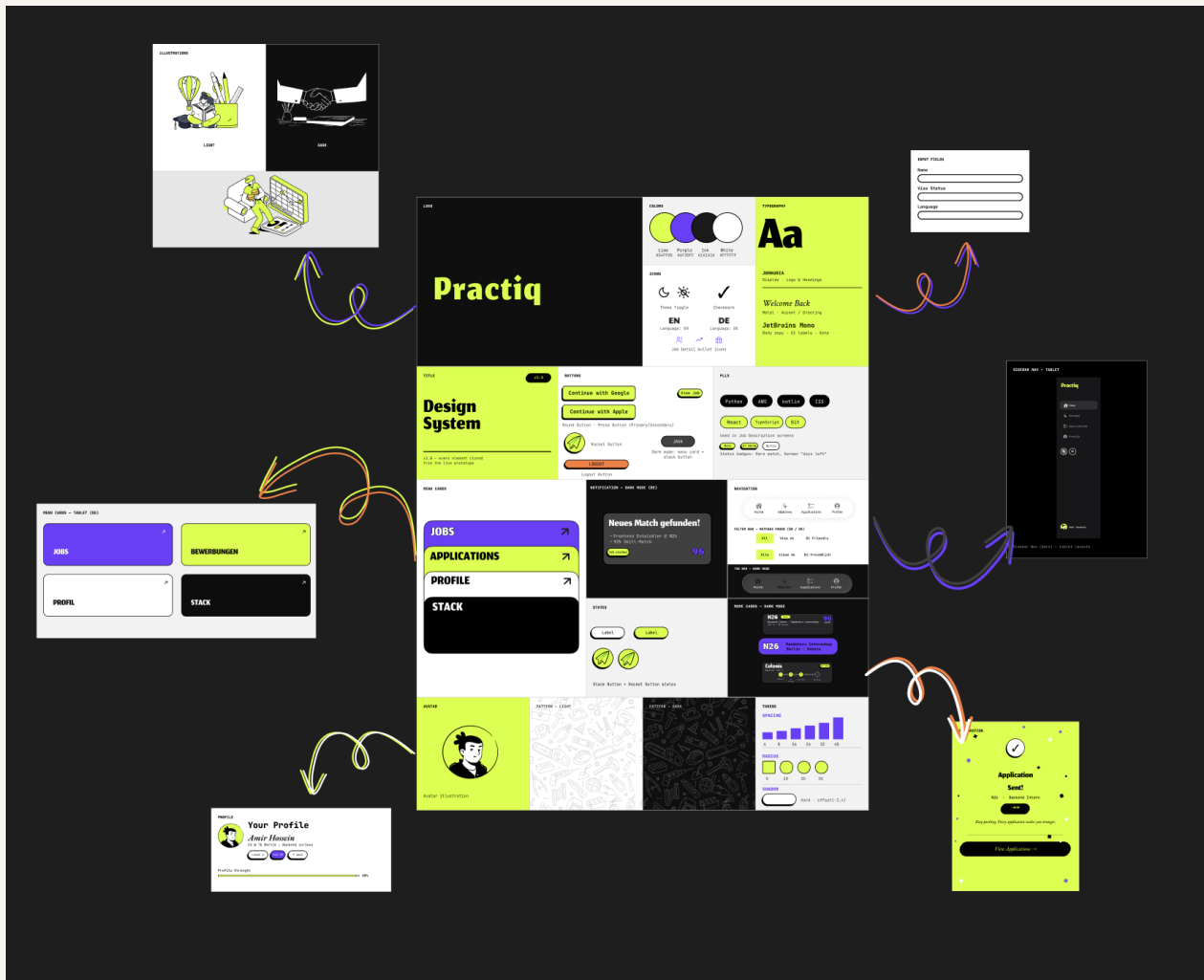
1.3.6 Prototype

The clickable prototype lives in Figma: 48 mobile frames (390×844, with the Matches list drawn taller so it scrolls) and 56 tablet frames (1366×1024), alongside the four sidebar frames and the component set. **Every core screen exists in four variants: Light / Dark × EN / DE.** I treated dark mode as a real design task rather than a colour inversion, which is how I tried to satisfy the professor's "light and shadow" brief. In light mode attention is directed by hard shadows dropping off white cards, with the shadow token set to a hard offset of $-3/4$. In dark mode there are no shadows to lean on, so attention is carried by the lime accent glowing on near-black Ink. In both modes the primary action stays the brightest object on the screen. The German build is a full translation: "Bewerbungen", "Profil", "Neues Match gefunden!", "Visum ok · B1-freundlich." The Apply moment is staged instead of being an instant screen swap: tapping APPLY floods the screen in lime, and only then does "Application Sent!" fade in with the +30 XP chip, so the reward lands at the exact moment Amir is most uncertain whether he did the right thing.

figma.com/design/xCXgrj9VWwUYwUpBD8PtTm/Practiq (in the folder as .fig and PDF export).

1.3.7 CI Guideline / Design System

The design system lives on its own Figma page, versioned v1.0, and every component on it was cloned out of the working prototype.



The v1.0 system page, shown with the satellite artboards it feeds — the illustration set in light and dark, the input fields, the tablet sidebar and the tablet menu cards in German, the profile header and the Celebration screen. The label printed on the sheet itself reads “every element cloned from the live prototype,” which is the order of work described above admitted on the artefact itself.

- **Colours.** Lime #D4FF00 for action, success and celebration, Purple #6F3DFF for the Jobs card, the two strongest match scores and the “Quest complete” badge, Ink #1A1A1A for dark surfaces and text, White #FFFFFF. The system runs on one rule I gave myself and kept: **lime is reserved for the action that moves Amir forward**: Apply, View Job, earned XP. Orange is the deliberate opposite and never marks a forward move. It is used twice in the whole app, on Logout and on the closing step of a quest that ended in a rejection, so that moving on and stopping can never be visually confused.
- **Typography.** Jomhuria carries the display, logo and headings, Metal, an italic serif, handles warm, human moments (“Welcome Back,” “Keep pushing.”), and JetBrains Mono carries body copy, UI labels and data. Three voices in one hierarchy: loud display, warm accent, precise data. The monospace choice is also a small signal to the developer audience that the app understands who it is talking to.
- **Components.** Google/Apple sign-in, primary and press buttons, the rocket submit, skill pills (Python, AWS, Kotlin, CSS, React, TypeScript, Git), status badges, the four colour-coded menu

cards, the match notification, the EN/DE filter bar, the bottom navigator, the tablet sidebar, the input fields (Name, Visa Status, Language), the avatar illustration set, and the light and dark background patterns.

- **Tokens.** Spacing 4 / 8 / 16 / 24 / 32 / 48, radius 5 / 20 / 30 / 50, shadow hard, offset(−3,4), and a dark-mode counterpart defined for every component.
- **Toggles as components.** The theme switch (moon/sun) and the language switch (EN/DE) are themselves reusable components, which is what makes the four variants of every screen maintainable.

2.0 Iterations

2.0.1 Change Log — what was iterated and why

The most important change is the first one, and the rest follow from it.

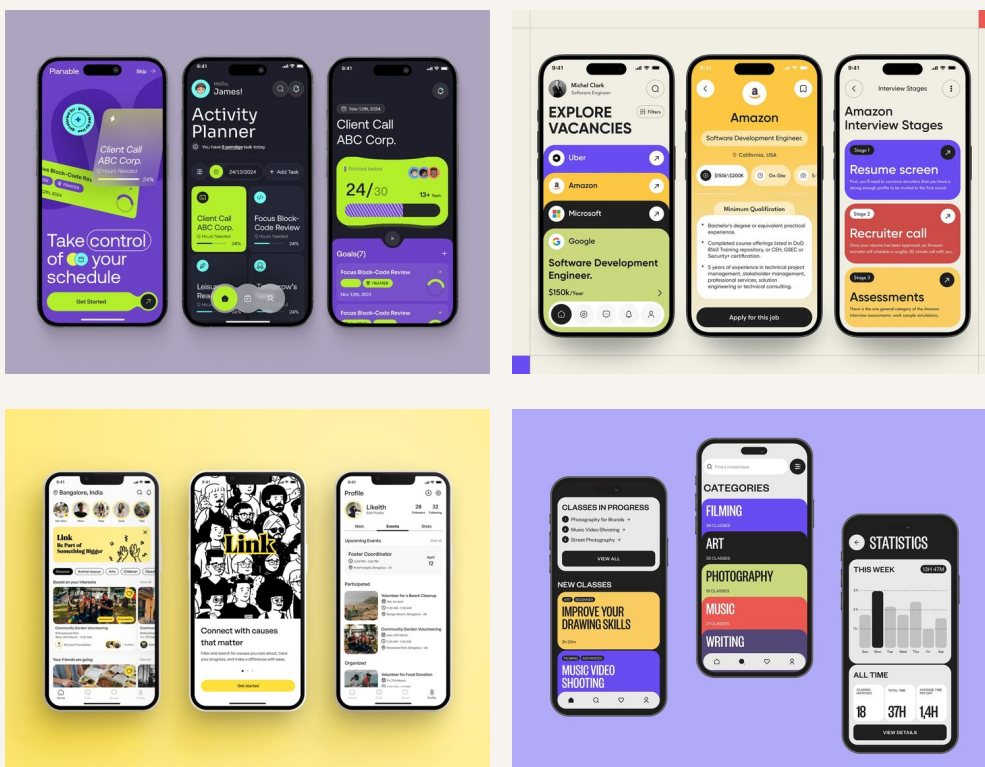
- **Plain dashboard → gamified quest system.** My Entry 1.1 concept was a neutral ranked dashboard (“12 Matches Found”). It was correct and it was lifeless. I kept the ranked list because the ranking genuinely helps, but I reframed the entire journey as quests: “Today’s quest” for matches, “Your quests” for applications, “Rare” badges, match scores, XP. Same underlying data, now carrying momentum as well as information.
- **Silent submit → animated Celebration.** The old flow ended at “Apply Now” and then simply moved on, which left a dead spot at the moment of highest doubt. The APPLY tap now triggers a staged animation, a lime wash and then “Application Sent! · +30 XP · Keep pushing. Every application makes you stronger.”
- **Added XP, Level, streak and Profile Strength.** A job hunt is normally invisible work: you apply, and then nothing happens for weeks. The XP layer lets effort itself accumulate visibly, so Amir’s sense of progress no longer depends on employers replying, which is the one thing he cannot control.
- **Reframed rejection.** In “Your quests,” a rejected application carries a “Quest complete” badge instead of a red failure state, and the stepper still shows how far he got, so a rejection becomes a finished attempt instead of a verdict on him. This was a late change.
- **Made the deadline visible, but gently.** Running applications show countdown badges (“4D left,” “9D left”). The deadline anxiety from the empathy map is acknowledged per application instead of looming as one vague dread over everything.
- **Added a full dark mode and a full German localisation** across every screen on both devices (the latter also satisfies Bonus Task I).
- **Replaced placeholder listings with seven real employers:** six rank in the match list (N26, Delivery Hero, Celonis, Trade Republic, Zalando, SAP) and Spotify appears in the tracker, so eligibility and status are demonstrated against realistic content instead of lorem ipsum.
- **Built a dedicated tablet version** with sidebar navigation, not a scaled-up phone.

2.0.2 Visual Evolution — six builds of the same product

The change log above is the argument and this section is the evidence. The Figma file still holds every generation of the six core screens side by side, and the sheets below are those generations in order, exported straight off the canvas. I have included them because watching the thing change from the first sheet to the last says more than any amount of prose about iterating could.

Build 00 — the reference board

Before any of the builds below there was a board. A lot of my visual input comes from saving interfaces on Pinterest, and in 0.2 B I argue that this only counts as research if I extract the rule and not the picture. This is that board, and this is me holding myself to it: four references, and for each one the single rule I took and the place in Practiq where it landed.



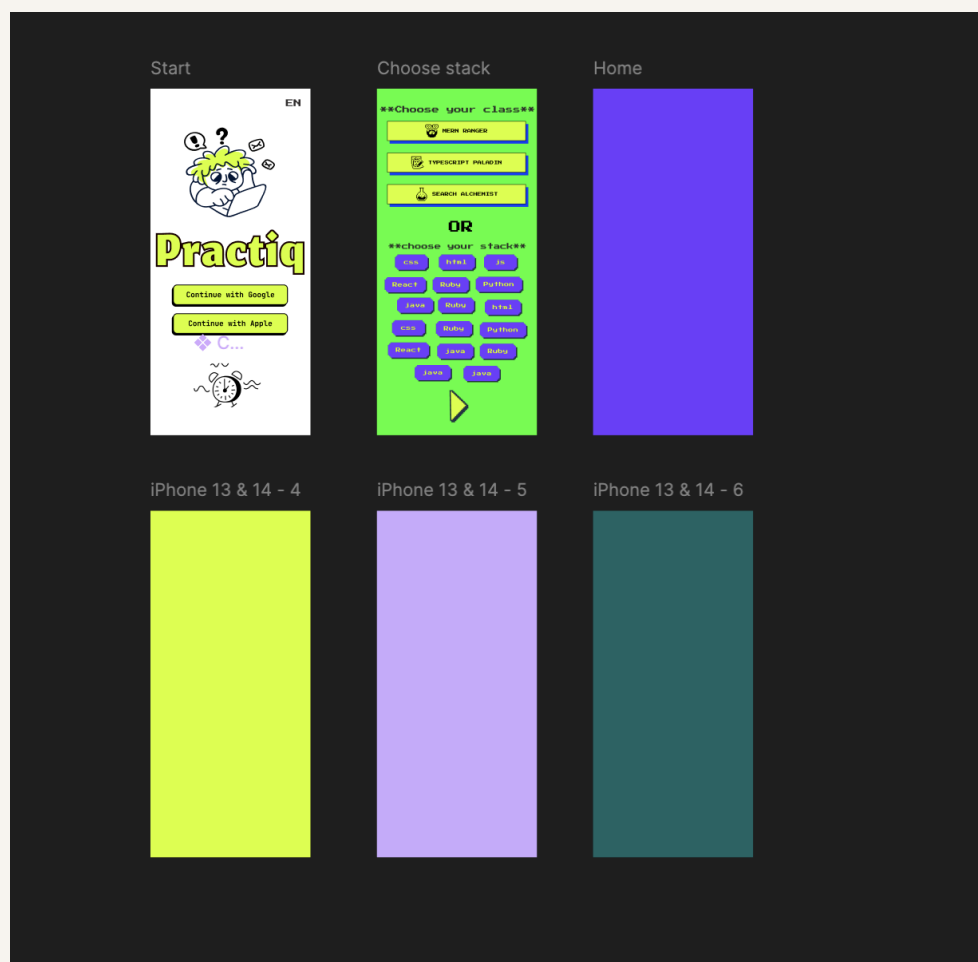
Reference board. Four interfaces collected on Pinterest during the visual research phase. All of them are other designers' work, reproduced here as citation rather than as my own, and they are read in the order the rules are listed below.

- **Planable, a task planner (CIPHERSLAB).** Acid lime and violet on near-black, oversized display numerals, and progress rings that make admin read as achievement. The palette started here. The rule I took was one saturated accent against one deep neutral, and it is the direct ancestor of lime on Ink in dark mode and of the XP, Level and Profile Strength block on Profile.
- **A recruitment app concept.** Colour-coded employer rows, a full-width dark apply button, and the interview process drawn as numbered coloured stages — Stage 1 Resume screen, Stage 2 Recruiter call, Stage 3 Assessments. The rule I took was that hiring is a sequence and should be drawn as one, which became the four-stage stepper in “Your quests”: Applied, In review, Interview, Decision.

- **Link, a volunteer coordination tool.** A single yellow accent, black line-art illustration used at full bleed as a surface, and warm second-person copy. The rule I took was that illustration can be the background instead of an ornament sitting on it, which is exactly the move I describe in Builds 05 and 06.
- **A brutalist interface set.** Heavy black outlines on every card, flat saturated blocks stacked as full-width bands, and condensed uppercase headings. The rule I took was to outline everything and not rely on shadow, which is why every card in the design system carries a border in both themes and why the palette survives dark mode without being reinvented.

What travelled from this board into the product is four sentences, not four pictures. The quest vocabulary, the eligibility score, the German build and the Deadline Companion have no ancestor here at all, because none of these references were trying to solve the problem in 1.2.1.

Build 01 — the arcade



Build 01. Three real frames and three colour studies — the point at which Practiq was still a game with a job board buried somewhere underneath it.

The first build was a 90s arcade game with a job app somewhere inside it. Onboarding did not ask Amir for his stack, it asked him to **choose your class**: MERN Ranger, Typescript Paladin, Search Alchemist, with the actual stack pills sitting underneath almost as an apology. The type was pixel monospace, the screens were full-bleed acid green, the buttons were purple, and every surface

was competing for attention at maximum volume.

Less flattering: only three of the six frames are real screens. The other three are flat rectangles of purple, lime, lilac and teal. I was auditioning accent colours before I had anything to put them on, which is the wrong order and the same error as drawing a design system before the product exists. I was choosing a palette against nothing.

The build died for one reason. It was themed rather than designed. “Choose your class” is funny once. It is irrelevant the second time somebody opens the app, and none of it was answering the FEELS quadrant from 1.2.3 at all. Amir is not looking for a joke. He wants to know whether he can apply, before the deadline moves again, and a retro costume does not tell him. Five colours shouting at once makes it harder to find out.

Build 02 — minimalism, and the palette starts collapsing



Build 02. The retro shell is gone, EN and DE are both present from the start, and purple is now doing the job acid green used to do.

The arcade shell came off in one pass, and what replaced it is the first version I would call designed and not just decorated. White space and type hierarchy carry the structure instead of colour, the pixel typeface is gone, and the screens settle into the six screen shape the product still has. Start, Choose stack, Home, Matches, Applications, Profile.

Two things start here and never stop. One is EN/DE parity, because this sheet already has both languages side by side, and that is the only reason the German did not end up bolted on and

translated at the end. The other is the illustration voice, the hand drawn line work that carries all the way through to the build I have now.

The colour is still wrong at this stage. Purple has replaced acid green as the dominant surface, but the idea of a dominant surface has not gone anywhere. A purple screen with lime buttons on it is perfectly legible. It is also a screen where the background is louder than the content sitting on top of it.

How the colour actually changed

The palette moved in three steps, and every step made it smaller.

- **Build 01 — colour as theme.** Acid green, purple, lime-yellow, lilac and teal. Five hues with no hierarchy between them, because colour was the personality of the app and so every screen wanted one of its own.
- **Build 02 — colour as surface.** Down to two, purple and lime. Better, but purple was still filling whole screens, which meant lime had to fight the background every time an action needed to be found.
- **Final — colour as meaning.** Purple no longer fills a screen, it marks one. Cream and Ink carry all of the structure, lime is reserved for the one action that moves Amir forward, and purple holds a small, specific set of meanings: the Jobs card, the two strongest match scores, and the “Quest complete” badge. Orange appears exactly twice in the whole app, on Logout and on the closing step of a rejected quest, so that stopping can never be visually confused with progress.

The rule I ended up with in 1.3.7 — lime is reserved for the action that moves Amir forward, and lime is never text — is one I could not have written during Build 01, because Build 01 had nothing to reserve it from. The palette only became defensible once it got smaller, which is the real reason the app ended up so much quieter than the concept.

The illustrations

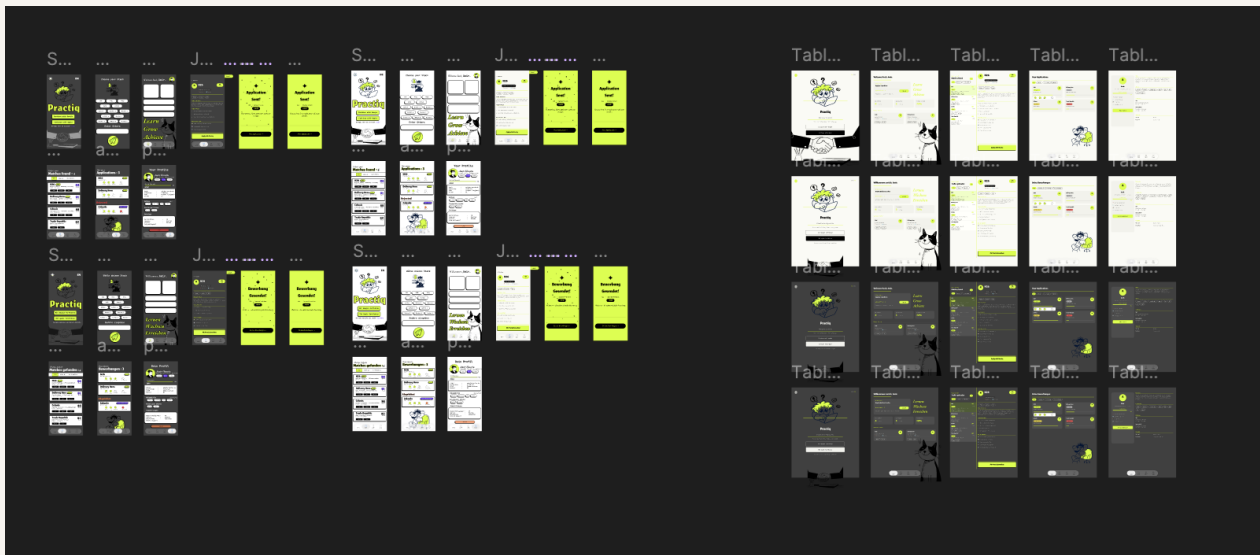
In Build 01 the illustration is a sticker. There is a doodle of a stressed student with lime hair, an alarm clock and a handshake, and they sit on the screen the way a decal sits on a laptop lid: present, likeable, and structurally doing nothing.

Turning them into a system took three decisions:

- **Fix the drawing rules.** One line weight, black line art, and lime as the only fill ever permitted inside a drawing. That single constraint is what makes the cat beside “Learn Grow Achieve”, the handshake under the sign-in buttons, the slumped character on Applications and the avatar on Profile all read as one hand.
- **Give every drawing a job.** The handshake sits under the sign-in buttons because that screen is about being let in. The cat sits beside the three-word promise because that line is the emotional argument of the whole app. The slumped character on Applications sits exactly where a rejection appears, which is the one place on the screen where a human touch is doing real work and not just being charming.
- **Let the drawings become the background.** This is visible in Builds 05 and 06 and it is the decision I am most pleased with. Once the line work was consistent enough to tile, illustration

stopped being something placed on top of a screen and became the surface the screen sits on. It carried an accessibility cost I had to design around, since the pattern has to be held far below the cards in contrast so that it always reads as texture and never as content. That is the Figure/Ground argument I make in 2.1, arrived at here by necessity and not borrowed from a book.

The tablet, and the mistake I made and then undid



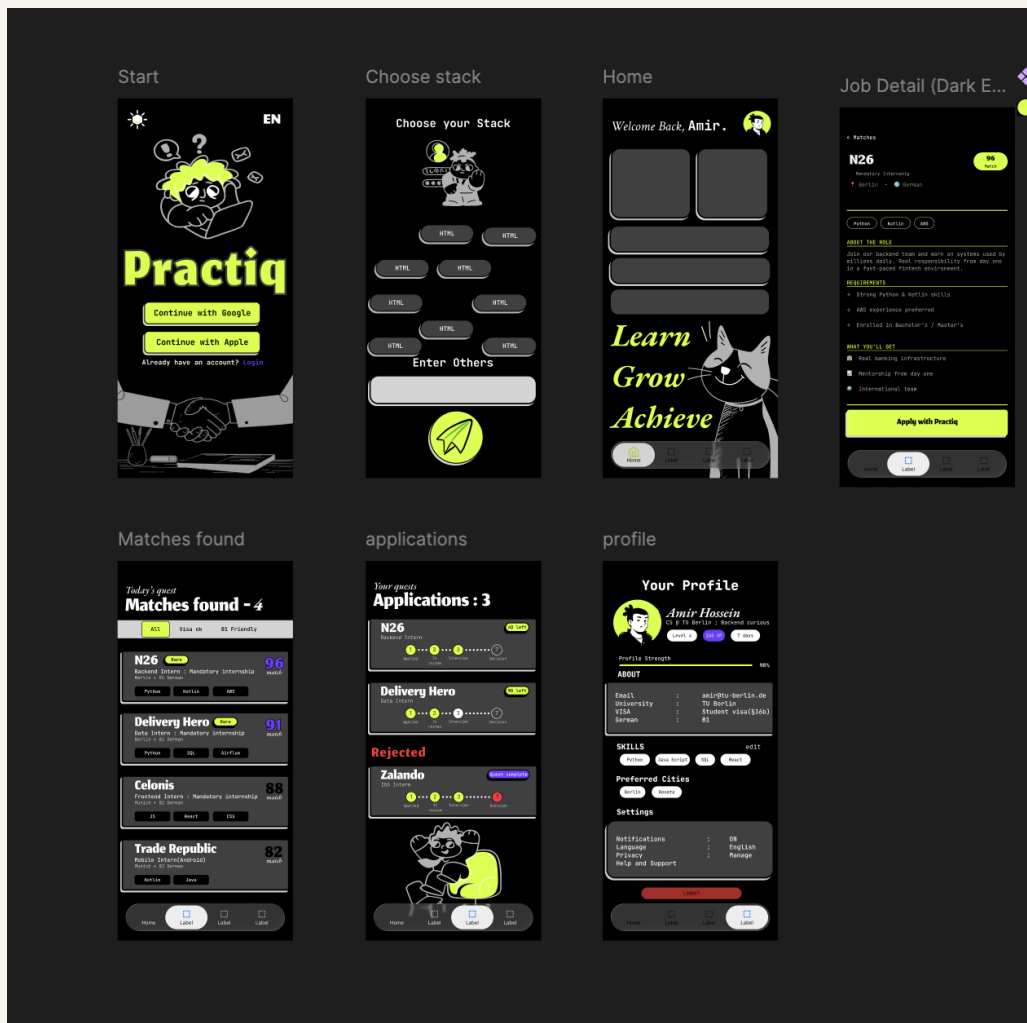
Build 03. The moment the tablet joined the file, sitting on the same canvas as the phone before it was pulled back out into a build of its own.

My first answer to the tablet was to do the whole thing in one screen. The reasoning was that a tablet has room, so everything the phone splits across separate screens could be folded onto a single dense surface and Amir would never need to navigate at all. Written down like that it still sounds like using the device properly.

It was wrong and I undid it. A screen with everything on it has no hierarchy, so “what do I look at first?” came straight back as soon as the screen got bigger, and that is the exact question the ranked match list is supposed to answer. It was also worse than that. Folding the screens together meant the phone and the tablet were not the same product any more. They were two products sharing a palette, and every change after that would have had to be designed twice in two different shapes and then kept in sync by hand.

So I pulled the tablet back apart into a separate build carrying the same screens in the same order as the phone, and solved the extra space with a side rail instead of a merge. That is the version described in 1.3.2, and it is why the tablet Home can afford to drop the rail for a two-by-two card grid with a Recent Activity feed underneath. Same screens in the same order, built at a different geometry. Jakob’s Law is the argument: Amir should not have to relearn the app because he picked up a larger device.

Build 04 — dark mode as a design task

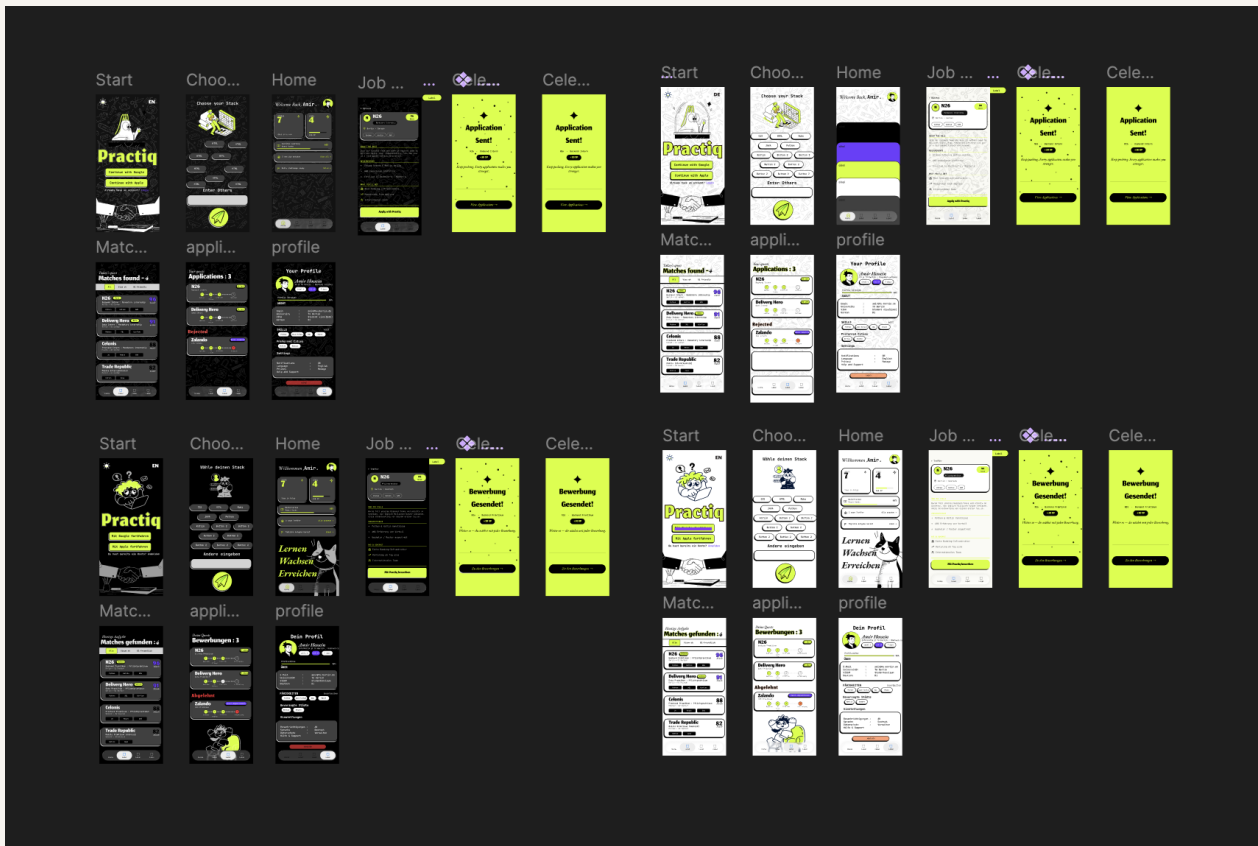


Build 04. The first sheet where the screens are legible at full size, and where Job Detail and the Celebration moment enter the product.

This is the build where dark mode stopped being a toggle in my head and turned into work. In light mode attention is directed by hard shadows dropping off white cards. Take the shadows away and that entire mechanism is gone, so the hierarchy has to be rebuilt from scratch on the lime accent glowing against near-black Ink.

Two screens the product could not do without arrive here as well. Job Detail answers Amir's fears in the order he actually feels them: employer, role, city and language, then the skill pills, and only then the description. The Celebration screen turns the moment after APPLY from a dead spot into the peak of the entire experience.

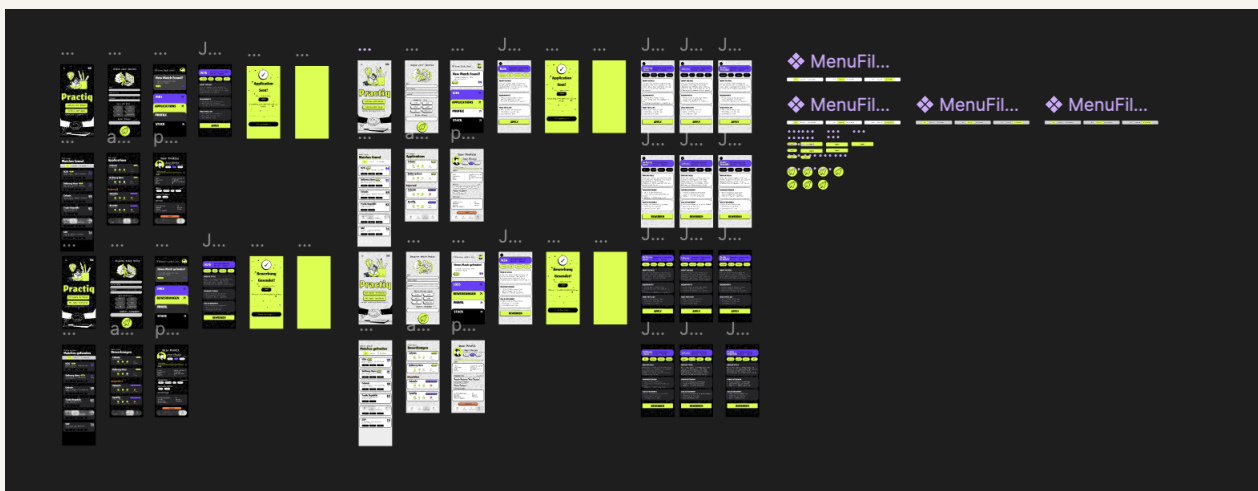
Build 05 — four variants, one system



Build 05. Light and dark against EN and DE, four complete sets carrying the same nine screens.

This sheet is where the arithmetic caught up with me. Four sets of the same screens is the point at which maintaining copies by hand stops being merely inefficient and starts being unsafe, and it is exactly where the components lesson from 0.2 B got learned the expensive way. The six tablet match cards still carrying identical skill pills are the receipt for that, and they are visible in this sheet if you know to look for them.

Build 06 — the prototype I have now

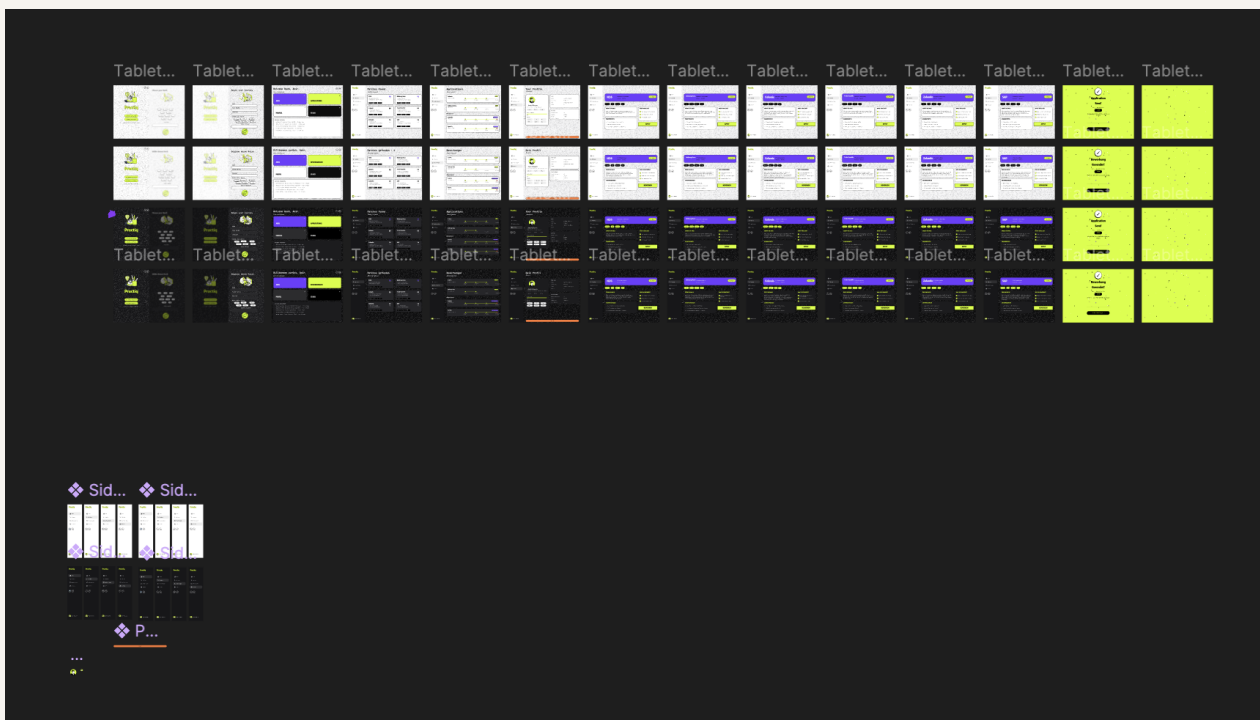


The finished mobile build. Four complete variants — Light and Dark against EN and DE — with the reusable filter bars, label

and button states sitting beside the screens on the right of the canvas rather than being redrawn inside them.

The last thing to change was Home. It had been a stack of unlabelled grey boxes for four builds and it finally turned into four colour coded destinations. Jobs in purple, Applications in lime, Profile in white and Stack in black. On the phone they are four labelled bars stacked vertically with an arrow on each one, and on the tablet the same four become a two by two grid. You can tell which one is which by colour before you have read the label, and for a user whose German reading speed is a stated barrier that solves a small piece of the language problem with colour instead of words.

The second thing this sheet shows is the fix to the problem I describe in 0.2 B. To the right of the screens there are component sets: the EN and DE filter bars, the label and rocket-button states, the match cards. The four variants are finally maintained from one source instead of by duplication. That is the components lesson arriving in the file and not only in my head, and it is the difference between a prototype that can be changed and one that can only be redrawn.



The finished tablet build — fourteen screens across four variants, which is exactly where the 56 tablet frames quoted in 1.3.6 come from. The sidebar navigator and the background pattern sit underneath as components rather than as copies.

This is the sheet I would point at to show that the re-layout argument in 1.3.2 is real rather than claimed. Fourteen screens run across every variant: Start, Begin your journey, Home, Matches, Applications, Profile, six per-employer Job Detail screens and the two Celebration frames. They are the same screens in the same order as the phone, so Amir is not being asked to relearn anything when he changes device. What differs is the geometry at every step: the split purple and black panel on Home, the wider match rows, the Job Detail header running across the top instead of stacking. The sidebar navigator that replaces the bottom bar is itself one component, not something redrawn fourteen times.

Everything else here is the argument of the previous five builds finally settling: one accent, one hard shadow, three type voices, the pattern held quiet behind the cards, and both languages carrying equal weight.

How the screen count grew

The count is the least glamorous evidence in this document and probably the most honest.

- **Wireframes:** five grey screens, no colour, no styling.
- **Build 01:** six frames, of which three were real screens and three were colour studies.
- **Build 02:** six screens, already doubled across EN and DE.
- **Build 03:** the tablet joins the file and the frame count roughly triples.
- **Builds 04–05:** nine screens per set once Job Detail and the Celebration moment exist, across four complete sets.
- **Build 06:** 48 mobile frames and 56 tablet frames, 104 in the file.

None of that growth was planned. Screens did not multiply because I wanted more screens, they multiplied because every honest answer to a design question costs frames: Amir needs to see eligibility before the description, the moment after APPLY cannot be empty, German cannot be half-done, a tablet is not a large phone. The number is what taught me why components exist. I could have read that lesson in an afternoon and instead I paid for it across a semester.

What the sequence taught me

Build 01 was the most fun I had all semester and the least useful thing I produced, and those two facts are related. It was designed for me. The six builds after it are the record of handing the app back to Amir one decision at a time: colour stopped being personality and became meaning, illustration moved from decoration to surface, the tablet stopped being a clever trick and turned into the same product at a different size, and the gamification survived only in the parts that answer a specific line in the empathy map. What did not survive is everything that was there because I enjoyed making it.

2.0.3 User Flow

Login → “Begin your journey” (stack, visa, language) → Home → Matches (ranked and filtered) → Job Detail → Apply → Celebration → Applications. The key efficiency move is that eligibility is collected once, at onboarding, so the system ranks and filters before Amir ever sees a listing. The interface does the cognitive work instead of the user.

2.1 Laws of UX and Gestalt Principles

I tried to use these as design tools and not as decoration, so for each one I name the exact place in Practiq it shows up and the empathy-map line it serves.

- **Goal-Gradient Effect.** Motivation rises as a goal comes into view, so “Your quests” draws each application as a stepper with filled dots, and the 98% Profile-Strength bar sits one nudge from full. Both target the “getting somewhere” job.
- **Peak-End Rule.** People remember an experience by its peak and its end, so I engineered the animated Celebration to be the peak. The memory Amir keeps of Practiq is “+30 XP, I’m stronger,” not “form submitted.”

- **Zeigarnik Effect.** Open loops stay in mind, so the 98% bar and the “4D left” quests are deliberately unfinished and clearly close, which pulls him back to close them.
- **Variable Reward.** Match quality changes from day to day (scores from 96 down to 74, occasional “Rare” badges), and that unpredictability is what makes opening the app feel worth it instead of feeling like a chore.
- **Hick’s Law.** More options mean slower decisions, so the filter tabs collapse “can I even apply?” into one tap and the ranking answers “which first?” before he asks.
- **Jakob’s Law.** Users carry expectations from other products, so match scores, XP, levels and streaks reuse the mental models of Duolingo and dating-app match percentages. The core loop needs no tutorial because Amir has met it before.
- **Von Restorff Effect.** The item that differs is the one remembered, so the lime element on a screen is the forward action. Orange is reserved for the two things that are pointedly not forward moves: the Logout button, and the closing step marker on a rejected quest.
- **Framing Effect.** The same fact framed differently lands differently, so a dead-end rejection is framed as “Quest complete.” This is my most direct answer to the THINKS quadrant’s “Am I competitive?”
- **Gestalt, Common Region.** Elements inside one boundary read as a group, so each match card is one bounded region holding everything needed to triage a listing: role, city, the German level it demands (Berlin + B1 German), the stack and the score. The visa question is answered one level up by the Visa ok filter, which is why the card never has to repeat it.
- **Gestalt, Similarity.** Shared shape reads as shared meaning: a pill is always a skill, a badge is always a status, a card is always a destination. This is readable by shape alone, which again softens the language barrier.
- **Gestalt, Figure/Ground.** The tool-pattern background is held at low contrast behind white cards in light mode and sinks into Ink in dark mode, so content always reads as the figure and never fights the ornament.

2.2 UX Writing

Before I styled anything I did a words-only pass of the core loop: a wireframe where every element is replaced by nothing but its copy, positioned where it actually sits on the screen. The idea was that if the words can carry the experience on their own here, with no colour and no layout to lean on, then the visual design only has to reinforce them later.

ONBOARDING	HOME	MATCHES	JOB DETAIL	APPLICATIONS
Begin your journey	Welcome Back, Amir.	Today's quest	N26	Your quests
Name	New Match Found!	Matches found — 4	Mandatory Internship	Applications : 3
Visa Status	Frontend Developer @	All · Visa ok · B1 Friendly	Berlin · Remote	Celonis — 4D left
Language	N26	N26 · Rare · 96	Visa ok · B1 German	Applied · In review ·
Choose your Stack	92% skill match	Delivery Hero · Rare · 91	Python · AWS · Kotlin ·	Interview · Decision
[Continue]	[View Job]	Celonis · 88	CSS	Delivery Hero — 9D left
	Jobs · Applications	Trade Republic · 82	About the Role	Zalando — Quest
	Profile · Stack		Requirements	complete
			What You'll Get	
			[Apply with Practiq]	
			Application Sent! · +30	
			XP	
			Keep pushing. Every	
			application makes you	
			stronger.	

Words-only pass of the core loop, laid out on the same five screens as the wireframe in 1.3.4 — copy only, with no colour, no layout and no device frame to lean on. Square brackets mark the single forward action on each screen.

The principles I held myself to:

- **Reassurance first, in the second person.** “Keep pushing. Every application makes you stronger” does work that “Application submitted successfully” never could. A neutral confirmation ignores the anxiety in the empathy map, and affirmed progress answers it.
- **State the answer, not the requirement.** “Visa ok · B1 Friendly” rather than “German B1 required.” Same fact, opposite emotional direction, and the positive framing is deliberate.
- **Name the user's job in his own words.** The Job Detail header says “Mandatory Internship” because “Pflichtpraktikum” is literally the word already in Amir's head, and it is an instant relevance signal.
- **Keep one metaphor.** “Today's quest,” “Your quests,” “Quest complete” are one consistent mental model, not three different words for matches, applications and outcomes.
- **Write the German, don't machine-translate it.** “Neues Match gefunden!” and “B1-freundlich” keep the same warm register as the English, which a literal translation would have flattened.

2.3 Accessibility Analysis (WCAG 2.2)

A lime-and-purple palette is exciting and, honestly, risky for contrast, so I audited it against WCAG 2.2 instead of assuming it passed.

- **1.4.3 Contrast (Minimum).** The two dangerous pairs are lime on white and purple on Ink. My rule is that lime is never used as text: it is either a surface with Ink text on top (Ink-on-Lime measures roughly 13:1, comfortably AAA) or an accent shape. White-on-Purple is only used at large, bold sizes where the 3:1 large-text threshold applies.
- **1.4.11 Non-text Contrast.** Pill borders, icons and the navigation indicator were checked at 3:1 or better in both themes, including the tricky lime element sitting on an Ink card.
- **2.5.8 Target Size (Minimum).** Navigation icons, filter tabs and match cards clear the 24×24 px minimum, and primary buttons are full-width on mobile.

- **1.4.1 Use of Colour.** Status is never carried by colour alone: every badge has a text label, and match strength shows a number (96) as well as a hue, which also protects colour-blind users.
- **1.4.4 Resize Text.** The layouts are auto-layout throughout, so text scales up without clipping.
- **3.1.2 Language of Parts.** The EN/DE toggle switches the whole interface at once. A half-translated screen is its own accessibility barrier, and for this exact target group it would be a cruel one.
- **Findings & fixes.** I had lime coloured text before and removed it after the analysis. I also thickened and darkened the navigation indicator's active-state underline so that it stays above 3:1 against both the light and the dark theme background.

2.4 User Testing

Setup: *[1 participant – Hesandi, 23, female, Sri Lankan national, studying Electrical and Information Engineering at HAW Hamburg, a daily mobile user, testing the prototype for the first time].* Tasks: onboard → find a compatible job → apply → find that application's status. I ran two back-to-back think-aloud sessions on the same build, one on the mobile prototype and one on the tablet prototype.

Recording: screen + audio for both sessions, referenced in HCI/user testing/ (mobile test audio.m4a + mobile test video.mp4; tab test audio.m4a + tab test video.mp4)

Findings: On mobile she moved unprompted from onboarding through apply to status and narrated the whole way: “this finds me the matching jobs, that's so cool”, “this was what I needed when I was trying to apply for my internships, I wish I had this one”, and on the status screen, “I can read my application here... this is so easy [compared to] reading a lot of emails.” On the profile tab she noted “it shows all my details so I don't have to edit every time when I'm logging in.” She found dark mode and the EN/DE toggle without being told they existed and reacted well to both, closing with “the navigation... is Apple-like... so minimal. Good job, I will definitely use this for my future applications.” On tablet, her first instinct was to check whether the layout was just a stretched phone view, and she confirmed it wasn't: “it's not a stretched tab version of the mobile one, it's exactly built for a tablet.” She read the match-percentage badge and used it to decide (“I would go with the 96”), called the left-hand navigation easy to use, re-checked dark mode and the compact profile card, and closed with “it feels like a finished product already... all buttons feel functional.”

Insights → changes: *[Neither session produced a task failure, a moment of confusion, or a request for help, so this round reads as validation rather than a fix list. The specific things she praised – a number on the match badge rather than colour alone, the tablet layout being a real second layout rather than a stretch, profile data persisting, EN/DE and light/dark being discoverable unprompted – map directly onto decisions already made in §2.1–2.3. The one item worth flagging is that she stumbled over the profile tab's label out loud (“my profilancy”), which is likely just a slip of the tongue but is a cheap copy check to re-run.]*

Reflection: *[A single-participant think-aloud is good at telling you a design is not actively broken and is a weak test of much else. Hesandi is a daily mobile user and knows the designer, so the near-total absence of friction is partly a sampling and social-desirability effect and not proof the flow holds up for a first-time, non-technical, or less digitally fluent user. Running both form factors back-to-back with the same person on the same day was useful for isolating device-specific reactions, since she spontaneously compared the tablet layout to the mobile one without being asked to, which a single-device test would have missed. It is also worth saying that she is not a computer science student and does not work in*

UI/UX, so this test gives me a non-technical point of view; a proficient UI/UX designer would almost certainly have caught more errors and mistakes than this round did.]

2.5 Ad

2.5.1 Google Text Ad

ad • practiq.app/students

Practiq — Internships Matched to Your Visa & Skills

Stop scrolling irrelevant listings. See only internships that fit your visa status and tech stack, ranked by match. Free for students.

The headline leads on eligibility filtering, which is the one thing LinkedIn, Indeed and StepStone do not do for this group. “Free for students” covers the cost objection.

2.5.2 Display Ad



A 9:16 story ad. It opens on the Celebration moment instead of a feature list, for the same reason the product does. The second screen covers the difference from a normal job board.

3.0 The Physical Prototype

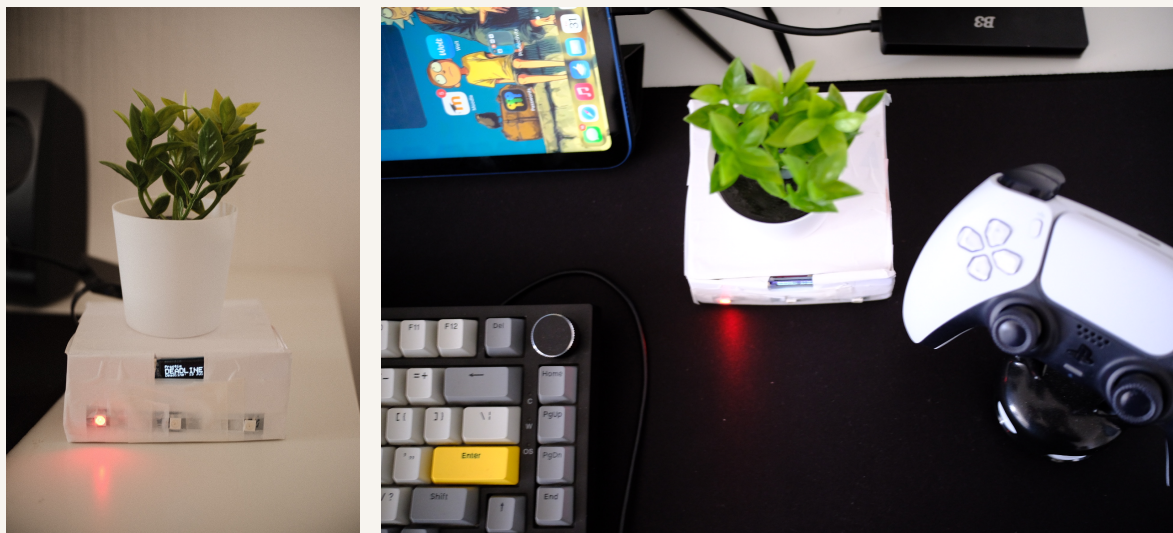
3.1 Concept — “The Deadline Companion”

The physical prototype takes Practiq’s progress mechanics off the screen and puts them onto Amir’s desk. My reasoning is that the deadline anxiety in the empathy map is ambient anxiety, the kind that does not switch off when the app closes. An object sitting on the desk can answer “am I getting anywhere?” at a glance and in his peripheral vision, without him having to open anything and drop back into the check-and-refresh loop.

Why a desk object at all. The form follows from where the work actually happens. Lectures aside, a student in Amir’s position does the studying, the coding, the CV rewriting and the applying from the same seat, in front of the same screen, for hours at a stretch, and that is an unusually reliable fact to design against — it is the thing that makes an ambient object viable here when it would not be for somebody moving around all day. If you know where a person will be sitting, you know where to put a signal.

It also makes the passivity the point. The device asks for nothing. It does not buzz, it does not wait to be tapped, and there is no moment at which Amir has to remember that it exists. He sits down to work on something else entirely and the deadline is already in the room with him, sitting at the edge of vision for the whole time he is heads-down on a problem. An app can only speak when it is opened or when it interrupts, and both of those are events Amir has to absorb. The object speaks by being there, and a phone cannot occupy that channel.

What I built is a small ESP32 device that makes the job deadline physically present. It is a low white enclosure with a planter seated into its top face, an OLED on the front carrying the countdown, and a short addressable LED strip along the front lower edge. The display reads *Practiq / DEADLINE / Deadline: 19 Jul*, and the strip moves along a green to amber to red gradient as that date approaches.



The Deadline Companion. The OLED carries the exact figure, the three addressable LEDs carry the approximate one, and the planter seated in the top face is what gives the object a reason to stay on the desk.

The reason for building this instead of adding another notification comes straight out of the

empathy map. Amir's problem is not that he is short of information. It is that the deadline drops out of his awareness entirely while he is heads down studying or writing code. A push notification is a thing competing with every other notification on his phone, and it gets swiped away in the same reflex as all the rest of them. An object that changes colour on the desk does not have to be checked at all. You read it the way you read a traffic light, peripherally and without ever deciding to look, and that moves a piece of Amir's cognitive load off him and onto the room.

3.2 Materials & Construction

- **Microcontroller.** An ESP32-S, chosen for the onboard WiFi, since the whole premise depends on the countdown being right without anyone maintaining it.
- **Display.** An SSD1306 OLED over I²C, wired to GPIO 21 and GPIO 22 for SDA and SCL.
- **Light.** A WS2812B addressable strip on GPIO 5 for data. Addressable rather than plain LEDs because the signal is a position along a gradient and not three fixed states, so each pixel needs its own colour.
- **Power and firmware.** Run at 3.3V, flashed from the Arduino IDE with the serial monitor at 115200 baud.
- **Time.** NTP over WiFi. The device holds the deadline date and derives days remaining from real clock time, so there is nothing to set and nothing that drifts.
- **Enclosure.** Hand-cut card, taped and painted white, with a circular aperture in the top face for the planter and two openings on the front, a slot for the OLED and a row of windows for the LEDs. It is still in progress and it looks it, shown unstaged.

The enclosure decision worth defending is that it is not a paper phone. It would have been easier to cut a rectangle and pretend it was a screen, and it would have proved nothing, because the whole claim is that this thing works differently from the app. The object needed its own form so that the light and the countdown are the interface instead of props inside a mock-up of a handset.

3.3 Ergonomics, Form, Affordances & Signifiers



In place. The device sits at desk level among the keyboard, the tablet and the controller stand, low enough to fall under the monitor's sightline and inside peripheral vision while Amir is working.

Size and placement. The footprint is roughly that of a paperback and the body is deliberately low, so it sits beneath the line of sight to the monitor instead of competing with it. It lives at the edge of the working area, next to the keyboard, which is the zone the eye crosses constantly without ever focusing on it.

The planter is the ergonomic argument. If the device only tells you about a deadline, sooner or later you turn it around or put it in a drawer. Putting a plant into the top face gives it a second reason to exist that has nothing to do with pressure. It earns its space on the desk as furniture, and the countdown comes with it. That is the difference between something you put up with and something you keep.

Signifiers. Colour does the signifying and text does the confirming. Green, amber and red is a borrowed convention that needs no instruction, which is Jakob's Law applied to an object instead of a screen, and it stays legible at a distance and at an angle where the OLED is not. The OLED is the precise read for the moment Amir actually turns and looks, and it carries the date and not a mood.

Affordances, and the deliberate absence of them. There is no button, no dial and nothing to touch. I left them out on purpose, because the moment the object accepts input it becomes another thing to manage and Amir already has an app for managing things. There is nothing here to press, only a state to notice, and that absence of affordance is what keeps it in the category of furniture.

Mapping. The signal travels along one axis with one meaning: as time runs down the colour runs from green toward red. It is worth naming that this is the opposite direction from the app, where XP and Profile Strength go up as Amir does more. The object tracks the clock and not his effort.

Those are two measurements that should not look the same, and mapping them identically would blur the exact distinction the device is meant to draw.

3.4 Proof of Concept



The peripheral read. From across the room the OLED is unreadable and the red is not — the object announces its state by the light it throws onto the white desk rather than by anything anyone has to focus on.

What drives the display. The ESP32 connects to WiFi on boot and synchronises with NTP. The firmware holds the internship application deadline as a fixed date, computes the days remaining against real clock time, prints that to the OLED, and maps the same number onto a colour along the strip. Nothing is set by hand and nothing needs resetting, which matters because a countdown that has to be maintained is a countdown that stops being true.

What it proves. The photograph above is the finding. At a distance the OLED is illegible and the red still is not, which is the ambient claim tested and not merely asserted: the state of the deadline is available without any act of checking. That is the one thing the digital prototype cannot do at all, because every piece of information it holds requires Amir to decide to open it first.

A gap worth stating. The device only knows the calendar. It does not read Practiq's state at all, so it has no idea how many applications are open or how far along they are. Closing that gap would need the app to expose its tracker state at an endpoint and the device to poll it on a timer. That is a small piece of work, and genuinely not done.

4.0 Presentation

4.1 Format & Structure (5–10 min)

The presentation is an HTML deck and not one made in slide software: fifteen keyboard-driven slides in the file `practiq-presentation.html`, typeset in the same system as the product, which makes the talk itself a small argument that the design language holds up outside the app. Arrow keys, the space bar, a swipe on a touch screen and a pair of previous and next buttons all move it, and a counter and a progress bar sit underneath so the room can see how much is left. It carries one claim from the title slide to the last one: a matcher that filters **before** Amir searches.

1. **Title.** The product vision in one sentence, with the core loop laid out as five steps — filter, match, apply, celebrate, track — so the shape of the whole talk is visible before it starts (30 s).
2. **01 — Naming.** Why the product is called Practiq: cut “-ikum,” paste “-iq,” so the name reads as Praktikum plus IQ. Praktikum names the goal and IQ names the method, which is applying with judgment instead of applying to everything (30 s).
3. **02 — Brand Validation.** The name checked against the three places it could quietly collide with someone else’s: the DPMA word-mark register, domain registries and social handles. Two real conflicts came back. `practiq.ch` already belongs to an unrelated Swiss healthcare company, and the `@practiq` handles on Instagram and LinkedIn are both taken. The formal DPMA query against Nice classes 9 and 42 has not been filed yet, and the slide says so instead of implying the name is clear (45 s).
4. **03 — Persona & Problem.** “Amir isn’t short on listings. He’s short on time.” The empathy-map quote runs under a mock of an unfiltered board where the listings read `visa unclear`, `requires C1` and `EU only`. The audience feels the problem before it sees a solution (45 s).
5. **04 — Design Principle.** “Filter once, at onboarding — not on every screen after,” broken into three moves: collect the context upfront, let the system do the elimination, and let every later screen inherit that trust. Mapped to Norman’s error prevention, with the onboarding screen shown beside it (45 s).
6. **05 — The Payoff.** “Rank the matches. Don’t list them.” A single compatibility score replaces reading every posting to judge fit, and Hick’s Law is named as the reason fewer, better choices are faster (45 s).
7. **06 — Trust Signals.** “Anxiety, answered first.” Visa acceptance and language requirement sit above the role description, grouped by Gestalt proximity, because they are the two facts Amir checks before anything else. Trade Republic runs as the counter-example, surfacing `Visa: Not supported` in the same glance (30 s).
8. **07 — Closing the Loop.** “The deadline stays visible, every session.” The tracker sits on the main path instead of in a menu, and the four-stage stepper lets him scan progress without parsing text (30 s).
9. **08 — Emotional Design.** The gamification layer argued and not merely shown: XP and Level replacing a blank checklist, quest framing carried in the vocabulary, and Kahneman’s Peak-End Rule deployed at the single most anxious moment in the app. The slide is explicit that a rejection still earns XP, because the moment most likely to end a session badly is exactly the one worth re-scoring (45 s).

10. **09 — Physical Prototype.** The Deadline Companion as a desk object rather than another notification: a small screen showing only the rarest, highest-match listing currently open and the days left to apply for it, and a three-light column tracking urgency across roughly a month, green to yellow to red. Mapped to Weiser’s Calm Technology and peripheral awareness (45 s).
11. **10 — Responsive Thinking.** “Same data. Different geometry.” The tablet is not the phone scaled up: tapping Continue shifts the dimmed left panel aside and brings Choose Stack into focus, so both states stay visible and context is never lost mid-flow. Jakob’s Law carries the consistency argument across breakpoints (45 s).
12. **11 — HCI Foundations.** Four principles located in the work, not listed: Norman’s affordances and signifiers, error prevention, Gestalt proximity, and the Peak-End Rule (30 s).
13. **12 — Colour System.** The slide I would defend hardest. Blue was the obvious choice and I rejected it deliberately, because blue is the colour of the bureaucracy Amir is trying to escape and it already means both “link” and “info” in most people’s heads. Lime is the only action colour, purple means “trust this number” so a statistic is never mistaken for something tappable, every card is outlined as well as shadowed, and the tokens are defined once in OKLCH so contrast stays predictable in both themes (45 s).
14. **13 — Bonus, shipped and not just mocked.** Every screen in the deck is also a running app: matches browsable, filters live, applications tracked, and state surviving a reload. This is the slide that turns the talk from a proposal into a demonstration (45 s).
15. **Closing.** “Less searching. More applying.” The constraint reframed as a filter and not a wall, handing over to the Deadline Companion (20 s).

4.2 Decisions I Am Ready to Defend

Why a ranked quest list and not a plain job board (Hick, Jakob), why an XP system belongs in a job app at all (the FEELS quadrant, answered through Goal-Gradient and Peak-End, with progress deliberately decoupled from employer response time), why a rejection reads “Quest complete”, the lime-reservation rule (Von Restorff plus the contrast strategy), why I used seven real employers instead of placeholders, and why the physical prototype complements the app instead of duplicating it.

5.0 Review

5.1 Expert Review of a Peer’s Project (ISO 9241)

What was reviewed, and how. The artefact is a peer’s prototype for a workplace English speaking-practice app, delivered as two clickable Figma prototypes, one mobile and one desktop, plus a design-system page. I reviewed it as an expert inspection and not as a user test: I walked every reachable path on both prototypes, recorded what worked and what did not, and judged the result against ISO 9241-11, which defines usability as effectiveness, efficiency and satisfaction in a specified context of use, and against the interaction principles of ISO 9241-110.

The course Excel template was not available to me, so the review is written out in full here instead of being attached as a filled sheet. The findings table below carries the same three columns that sheet asks for — the finding, the principle it offends, and a severity — so nothing is lost by the change of format.

Context of use (ISO 9241-11)

The specified users are working professionals and job seekers rehearsing English for specific workplace situations, not learners of general conversational English. The prototype names one, “Mia,” and labels her Intermediate. Their goal is competence in a named moment rather than fluency in the abstract: negotiating a salary, opening with an icebreaker, addressing a weakness, closing an interview. The context of use is private and spoken, since the primary input is the user’s own voice, and it spans two device classes with genuinely different postures. Everything below is judged against that context and not against a general-purpose language app.

Effectiveness

The core loop — choose a scenario, speak a timed round, receive a score, retry — is the right loop for the problem, and both prototypes carry a user from Home to a scored result without a dead end in the primary path. On mobile that path is Home, scenario list, scenario content, practice, results, with a working “Try Again”, plus Profile, Progress and a light and dark toggle wired in both directions. Desktop reaches the same outcome through its own flow.

Effectiveness is nonetheless the dimension with the most serious defects, because three things stop a user completing what the interface offers. “Next Round” on the results screen is not wired on either platform, which means the habit loop the whole product is built around cannot actually be repeated. On mobile only the Job Interview category is connected, so Warm Up, Team Meeting and Professional Writing present themselves as available when they are not. And the back control on mobile Results returns to the scenario description, silently discarding a completed three-round session with no warning and no way back to the result.

Efficiency

Efficiency is generally good and in places better than the brief required. Putting the microphone at the centre of the practice screen matches the fact that speaking is the whole interaction. The conversation preview and the phrasing tips — replacing “want” with “target”, “expect” or “seek” — do real work before a session starts, because they let a user judge whether a scenario is worth committing to instead of discovering it mid-round.

The desktop version is a genuine re-layout rather than a stretched phone, and it costs the user fewer steps for it: a persistent sidebar navigator, Home merged with Scenarios, and a two-column content screen that removes a level of drilling. The one efficiency cost is the Beginner, Intermediate and Advanced filter pills on mobile, which sit below a comfortable touch target and will produce repeated mis-taps on the control a user reaches for first.

Satisfaction

The visual work supports satisfaction well. The layout is calm and card-based, the teal identity holds across all three surfaces, and dark mode is a considered ink and not an inverted palette,

which is a distinction most student prototypes do not make.

The thing that undermines satisfaction is not the visual design. It is the consistency of the content inside it. The numbers and labels do not agree between the two versions, and somebody who opens both will not know which one to believe. Top Score reads 90% on mobile and 85% on desktop. “Team Meeting” turns into “Meeting Prep”. “Sessions” shows up elsewhere as “Scenarios Completed” and as “Lessons Completed”. The desktop scenario description repeats a passage in the middle of a paragraph. Each one of these is small on its own. Together they make the two prototypes feel like two different products.

There is also a missing piece of feedback that belongs here as much as anywhere: tapping the microphone produces no recording state. Nothing on screen confirms that the single most important action in the app has been registered, which is the point at which a user is most likely to lose confidence in it.

Where the design already conforms

An expert review should record conformance and not only defects, and three things are worth naming.

- **Controllability, handled unusually well.** The desktop sidebar’s “Dark Mode” item switches to the dark variant of whatever screen the user is currently on, instead of returning them to a fixed start screen. Preserving context across a theme change is a detail that is frequently got wrong and was got right here.
- **Suitability for the task, at the concept level.** Choosing workplace English over generic speaking practice makes every scenario legible at a glance, and the scenarios chosen are the ones that actually carry consequences.
- **Self-descriptiveness before commitment.** The conversation preview and the tips tell a user what a session will contain before it starts.

Findings against the ISO 9241-110 interaction principles

Severity is recorded on three levels. **Critical** means a user cannot complete an offered task. **Major** means the task can be completed but with avoidable loss, confusion or rework. **Minor** means the defect is noticeable but does not obstruct the task.

Finding	Principle	Severity
“Next Round” is not wired on either platform, so the core loop cannot be repeated	Suitability for the user’s tasks	Critical
Warm Up, Team Meeting and Professional Writing are presented but not connected on mobile	Suitability for the user’s tasks	Critical
Back from Results discards a finished three-round session without warning	Use error robustness, Controllability	Major
No recording state after the microphone is tapped	Self-descriptiveness	Major
No onboarding or competency assessment, so “Intermediate” and “Customized by you” are unexplained on a first run	Self-descriptiveness, Learnability	Major
Top Score reads 90% on mobile and 85% on desktop	Conformity with user expectations	Major
“Team Meeting” on mobile is “Meeting Prep” on desktop	Conformity with user expectations	Major
“Sessions” appears as “Scenarios Completed” and as “Lessons Completed”	Conformity with user expectations	Major
Difficulty filter pills fall below a comfortable touch target	Suitability for the user’s tasks	Major
The moon icon does not change after switching to dark mode	Self-descriptiveness	Minor
The desktop scenario description repeats a passage mid-paragraph	Self-descriptiveness	Minor
“Warm Up” and “Warm up”, “📧 Emails” and “📧 Email”, “Round 1 Complete” and “Round 1 Completed”	Conformity with user expectations	Minor

Craft and file hygiene

Two observations sit outside ISO 9241 because they are not usability defects, but they affect how the file is handed on. The primary colour swatch is written as #11AA698, which is seven digits and therefore not a valid hex value. And several file and layer names carry typos — Mobile verion, Scrol Bar, Beginer Button, Ice Braecker, Email Writting. Neither costs a user anything. Both cost the next person who opens the file.

The three changes I would make first

1. **Wire “Next Round”.** It is the natural next step after finishing a round and it is the app’s habit loop. Connecting it even just back to the practice screen matters more than any other single link in the file.
2. **Reconcile the numbers and labels across mobile and desktop.** The mismatches are almost certainly leftovers from iterating on one version without updating the other, and fixing them is what makes the two prototypes read as one product.
3. **Turn the design system into components the screens are actually built from.** The system page is well templated and reuses real content from the UI, which is most of the work. Making the screens instance those components instead of copying them is what would stop the mobile and desktop divergence above from happening again.

Overall assessment

Measured against ISO 9241-11 this is one of the more complete prototypes I reviewed. It is effective along its primary path on two device classes, it is efficient in the places that matter most to a spoken interface, and it is satisfying to look at and move through. It carries a real design system with four categories, a dark mode that was designed rather than inverted, and a desktop version that was rethought rather than stretched. The defects are concentrated in unfinished wiring and in content that drifted between two versions, and both are the kind of problem that is cheap to fix and expensive to leave.

5.2 Self-Review

Effectiveness. The core loop resolves the discovery-friction Job To Be Done, and the deadline anxiety is addressed on two fronts, digitally through the tracker and XP, and physically through the Deadline Companion. Measured against my Entry 1.1 concept this is a real gain, since the earlier version could tell Amir what existed but did nothing about how he felt.

Efficiency. I claim that a pre-filtered, ranked quest list cuts decision time compared with an unfiltered board. That is still an assumption until the usability test in 2.4 confirms it, an argument rather than a measurement, and naming the gap is part of doing this properly.

Satisfaction. The gamification layer is a designed response to the FEELS quadrant and not ornament, but its real effect on satisfaction is exactly the thing the user test has to verify and not something I can assert. I am also aware of the risk in it. XP and streaks can slide from motivating into trivialising something that is genuinely stressful, so the copy (“Keep pushing...”) is tuned to encourage, and I was careful never to gamify a rejection itself.

Process. The empathy map repaid its cost many times over and nearly every decision in this document traces back to it. If I did one thing wrong, it was testing too late, after the visual design was largely locked, which meant the test could reshape copy and micro-interactions but not the deeper information architecture. That is the first thing I would change next time.

6.0 Bonus Tasks & Appendix

6.1 Web Application (Bonus II)

The web build exists and is deployed:

`practiq-sheh4n-com-lsiv.vercel.app`

It was scaffolded with Lovable and then developed on from that repository with Claude Code until it matched the design system in 1.3.7, and what that workflow did and did not teach me is discussed in 0.2 B. It is not the phone layout stretched to fit a browser. `/home` is a landing page the app never had — a hero (“Learn, Grow, Achieve”), a search field, a three-step “How it works” strip, a statistics band (4k+ students matched, 120 partner companies, a 73% interview rate) and a preview of the top three matches — and only then do `/matches`, `/applications` and `/profile` carry the product itself. Navigation is the third answer I have given to the same question: a bottom bar on the phone,

a side rail on the tablet, and on the web a full-screen overlay drawer listing the six destinations numbered 01–06.

The change that matters is that the web version is **actually stateful**, and the Figma file can only be clicked through. In “Your quests” an ADVANCE button walks an application along the real stepper (Applied → In review → Interview → Decision), a red cross rejects it into the “Quest complete” group, and REMOVE deletes it. A job already in the tracker comes back marked APPLIED on the matches list, and the whole state, together with the theme and language choice, survives a reload in local storage. This is the first point in the project where the tracker described in 1.3.2 is a mechanism rather than a picture of one.

Both toggles from the design system survive and compose: EN/DE and Light/Dark. The German is a full translation again and not a veneer — “Deine Quests”, “Bewerbungen”, “Aktiv” and “Abgelehnt”, “4T übrig”, the stepper as “Beworben”, “In Prüfung”, “Interview”, “Entscheidung”, and “Quest abgeschlossen” for the reframed rejection. The layout is genuinely responsive (breakpoint utilities, a 600 px query and a coarse-pointer query) and it honours prefers-reduced-motion, which matters because this build leans on motion far harder than the prototype does.

Two gaps remain against the Figma prototype. The web matches list carries four listings (96, 91, 88, 82) and not six, and the “Rare” badge is replaced by an APPLIED state badge. It does one thing better than either device build: the match card answers the visa question on the card itself (Berlin · B1 German · Visa ok). That is what I claim in 2.1 a card should do, and neither the phone nor the tablet card manages it.

Bonus I (two languages) is delivered: every mobile and tablet screen exists in both EN and DE, and so does the web build.

Appendix

- A. Figma (.fig + PDF export): figma.com/design/xCXgrj9VWwUYwUpBD8PtTm/Practiq
- B. Physical prototype photos *[can find in the folder]*
- C. References (archived): lawsofux.com · w3.org/TR/WCAG22 · [ISO 9241-11](https://iso.org/iso9241-11) · sinus-institut.de · hubspot.com/make-my-persona · uxpressia.com · app.sistrix.com/de/serp-snippet-generator